Quotative inversion as smuggling: Evidence from Setswana and English

John David Storment
Stony Brook University

Abstract: Quotative inversion (QI) clauses in Setswana and English – two SVO languages – show a marked OVS word order in which a quote appears preverbally and the agentive external argument appears postverbally. In this paper, I present empirical evidence from both languages showing that the preverbal quotative subject moves to its landing site via A-movement, while the agentive subject remains in-situ. While moving an internal argument to an A-position above an external argument may seem to pose a minimality problem, I show that QI constructions are best conceived of as smuggling derivations in which the internal argument is moved over the external argument via the movement of a larger constituent containing the internal argument (Collins 2005; 2024; Storment 2025). I support this proposal empirically and theoretically, drawing on novel data from both English and Setswana analyzed under the Strong Minimalist Thesis (Chomsky et al. 2023). I additionally draw parallels to other inversion phenomena in both languages, showing that these constructions are best conceived of as inverse voice phenomena.

Keywords: smuggling, quotation, inversion, inverse voice, quotative inversion, voice, English, Setswana, Bantu, syntax

1 Introduction

Quotative inversion (henceforth QI) is a marked construction that exists in several languages with a non-canonical word order in which a direct speech report (or quote) verbal complement appears preverbally, and the thematic subject AGENT appears postverbally. QI is a productive construction for predicates of direct speech in English (Collins & Branigan 1997; Collins 2003; Bruening 2016), French (Doeleman 1998), Spanish (Suñer 2000; Etxepare 2008), Hungarian (Gärtner & Gyuris 2014), and Zulu (Buell 2006; 2007), among other languages, and has received relatively little descriptive and analytic attention, despite showing many interesting properties. QI is exemplified below in English, and in Setswana (Southern Bantu; ~8.2 million speakers).

(1) a. ‘Peace!’ said Saruman.
b. ‘Who is he?’ asked Frodo.
c. ‘Maybe not,’ answered Gandalf.

(2) ‘Heelang! Heelang! Tshaba-ng mo efatshe-ng la bokone,’ (Bible)
‘hey! hey! flee -IMP 18LOC land -LOC of north’
go bua Morena.[1]
SM17 say Lord
‘“Up! Up! Flee from the land of the north,” declares the Lord.’

In this paper, focusing on QI in English and Setswana – a language for which QI has not yet been described – I show that the quotative argument undergoes A-movement to its preverbal position, which I identify as Spec,TP. I additionally show that, in both languages, the thematic subject remains in-situ (Spec,vP) in QI constructions. Initially, these facts seem to present an

[1] I follow standard conventions of the Leipzig Glossing Rules for examples throughout the paper with the following additions: CL noun class, SM subject marker.

issue of Relativized Minimality (Rizzi 1990) – moving a lower argument to a position above a higher one. Crucially, I demonstrate that the movement of the quote to the subject position in both languages is facilitated by *smuggling*, an operation in which the quote is moved within a larger constituent (e.g., VP) to a position above the thematic subject in order to become a viable goal for A-movement (to Spec,TP). Additionally, I defend a conception of smuggling as a core component of grammatical voice alternations (Collins 2024; Storment 2025), thereby casting QI in both English and Setswana as a kind of inverse voice construction, a position for which I provide additional empirical support by showing that other kinds of inversion constructions in these languages, such as locative inversion, behave in a uniform manner.

All English data shown in this paper reflect the native speaker judgments of the author, and are additionally corroborated by naturally-occurring native speaker data collected from Internet searches, which offer a valid sample of spontaneous linguistic utterances and the underlying grammatical system of the authors of these utterances (see Fernández-Serrano 2022). The majority of the Setswana data comes from elicitation sessions carried out between the author and a native speaker consultant from February to June 2023. Additional Setswana data come from publicly available documents written in Setswana (e.g., the *Bible*), as well as additional field notes from other researchers (Collins 1997); these examples are indicated throughout the text.

In section 2, I show that both English and Setswana QI clauses involve phrasal movement of the VP containing the main verb and the internal argument THEME to a position above the thematic subject. In section 3, I demonstrate that the internal argument THEME comes to occupy the position of structural subject—Spec,TP—in QI clauses, and that the external argument Agent remains in its base position; I additionally elucidate on the category and structure of the quotative argument THEME. I discuss in section 4 how these data support an analysis of QI as smuggling for both languages, and I tie this fact to a novel conceptualization of grammatical voice. In section 5, I discuss the Transitivity Constraint of QI clauses (Collins & Branigan 1997), and show how it naturally follows from a smuggling account of QI. Section 6 discusses the similarities of QI to locative inversion in both English and Bantu, and I show how the technology presented here to analyze QI naturally extends to locative inversion. Section 7 concludes.

2 VP phrasal movement in QI clauses

There is somewhat of an open question regarding how the verb comes to linearly precede an external argument AGENT in inverted OVS-type clauses across SVO languages, especially in languages which lack v-to-T raising (see Collins & Branigan 1997, Collins 2003, Storment 2025 for discussions of this issue for QI, and Bailyn 2004 for the same question involving OVS word orders in Russian). Derivations in which the verb precedes an external argument AGENT in canonically SVO/SOV languages that lack v-to-T raising are troubling given more-or-less standard assumptions about the basic structure of a clause (e.g., that external arguments are introduced in Spec,vP and the verb does not move above v^0). Thus, the analytical question is how to get the verb to appear before the external argument, given that this is not normally permitted, without significantly complicating our notions of argument structure and verb movement.

I believe that these kinds of issues have been the motivation for certain analyses of inverted configurations such as locative inversion (Bresnan 1994; Culicover & Levine 2001; Doggett 2004; Rizzi & Shlonksy 2006; a.o.) or Russian OVS clauses (Titov 2012; 2013; Kasenov 2025) that claim that the postverbal thematic subject is always an internal argument, thus allowing for the verb to naturally move above it, even though such analyses complicate any notion of consistency in argument structure and theta-role assignment.

The natural alternative to these approaches seems to be to complicate the theory of verb movement, such as claiming that v-to-T movement is allowed in certain configurations in a language that otherwise lacks it (again, see Bailyn 2004). In past work, English QI constructions have been argued to involve head movement of the verb to its position preceding the external argument AGENT (e.g., Collins & Branigan 1997). In this section, I systematically show this analysis of verb movement cannot be correct for QI clauses in English and Setswana, and that it is further unnecessary to complicate the theories of argument structure and verb movement that we use to explain these kinds of inverted configurations.

In this section, I present numerous data from English and Setswana demonstrating that the VS (3)-(4) word order in QI clauses in these SVO languages (5)-(6) is derived by phrasal movement of the VP to a position above the thematic external argument (and I show in the following section that this external argument remains in its base position).

(3) 'Sméagol has to take what's given to him,' answered Gollum.

(4) 'Le kae? ga botsa Seabelo.
'and where?' SM17.PST ask Seabelo
'"How are you?" asks Seabelo.'

(5) Gollum answered, 'Sméagol has to take what's given to him.'

(6) Seabelo a botsa, 'Le kae?'
Seabelo SM.3SG.PST ask 'and where?'
'Seabelo asks, "How are you?"'

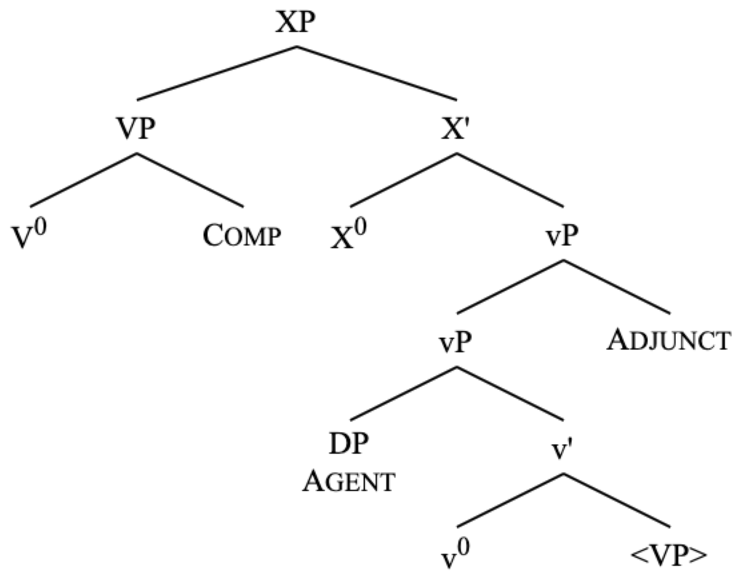
While it is possible to have a preposed quote without the verb appearing preverbally via Ā-movement (7)-(8) (see section 3.2), QI clauses are marked by the verb necessarily appearing before the AGENT as well. Assuming that the position of the verb in QI cannot be derived by some application of head movement that is not normally possible in the language, movement of the entire VP is needed to derive the word order of QI clauses.

(7) 'Sméagol has to take what's given to him,' Gollum answered.

(8) 'Le kae?' Seabelo a botsa.
'and where?' Seabelo SM.3SG.PST ask
'"How are you?" Seabelo asks.'

In this section, I show data from English and Setswana that isolate the VP as the element that moves above the AGENT. These data include the distribution of adjuncts (2.1) and control clauses (2.2) in English and Setswana, and the distribution of prepositional particles in English (2.3). The generalizations to draw from these examples are that VP-internal elements (such as control clauses—which are VP complements—and English particles) must appear in a position before the AGENT in QI, while adjunct material which is external to the VP (such as secondary depictive predicates, agentive adverbs, and purpose clauses) must follow the AGENT. These considerations motivate the following structure involving VP movement to a position above the external argument, which will be expounded upon in following sections.

(9) VP movement to a position above the external argument AGENT.



Many of the following tests appear in Collins (2003) and chapter 4 of Storment (2025), though only for English. The Setswana data presented here are a novel contribution, along with their parallels drawn to English. These data confirm that QI clauses warrant a similar analysis in these two unrelated languages.

2.1 Adjuncts

Adjuncts that are right-adjoined at the vP level, such as secondary depictive predicates, purpose clauses, and adverbs of manner, cannot intervene between the verb and the postverbal AGENT in QI clauses. These data show that nothing higher than the level of VP may be moved up in QI.

2.1.1 Secondary depictive predicates

Secondary depictive predicates which modify the AGENT of a predicate are typically taken to be adjoined to the projection that introduces the AGENT: the vP (Williams 1980; Mallén 1991; Pylkkänen 2008; Collins 2024). In both English and Setswana, AGENT-oriented secondary depictive predicates are permitted in QI clauses.

(10) ‘D’oh!’ said the man drunk.

(11) ‘Dumela!’ go bua Seabelo a tagilwe.
‘hello!’ SM17 said Seabelo SM1 drunk
‘“Hello!” said Seabelo drunk.’

Crucially, these secondary depictive predicates must follow the AGENT. They are not grammatical in a position preceding the external argument.

(12) *'D'oh!' said drunk the man.

(13) *'Dumela!' go bua a tagilwe Seabelo.
'hello!' SM17 said SM1 drunk Seabelo
int: '"Hello!" said drunk Seabelo.'

It should be noted that an English example such as (25) can improve somewhat with Heavy-NP Shift (HNPS) (Collins & Branigan 1997; Storment 2025). This is true of the other vP-level adjuncts discussed in this paper, though I do not show examples for all of them.

(14) ?'D'oh!' shouted drunk the man I've never met before.

The fact remains, though, that adjuncts such as secondary depictive predicates cannot ever precede the thematic subject in a canonical QI clause without HNPS or a strong contrastive focus emphasis on the AGENT. Following Den Dikken (1995), I assume that HNPS involves additional movement of the thematic subject in these cases that allows it to appear to the right of the secondary predicate. It should also be noted that HNPS-derived word order seems to be optional to the extent that (14) is of more-or-less equal status with the secondary predicate following the heavy thematic subject. The interactions of inversion and HNPS are interesting – and have received notably more attention in the literature on locative inversion (Culicover & Levine 2001; Rizzi & Shlonsky 2006; Diercks 2017) – however, I do not pursue them here in significant detail. An account of HNPS in locative inversion and QI can be found in Storment (2025).

As such, in both Setswana and English, the vP-level adjunct secondary depictive predicates can only appear adjacent to the verb and preceding the thematic subject in standard QI clauses.

2.1.2 Purpose clause adjuncts

Purpose clause adjuncts, another vP-level adjunct, must also follow the thematic subject in QI clauses. This fact is consistent in English and in Setswana.

(15) 'Fire!' cried Cecil to get the crowd's attention.

(16) *'Fire!' cried to get the crowd's attention Cecil.

(17) 'Molelo!' go go -ile Seabelo go kopa thuso.
'fire!' SM17 yell-PRF Seabelo SM17 plead help.
'"Fire!" has yelled Seabelo to call for help.'

(18) *'Molelo!' go go -ile go kopa thuso Seabelo.
'fire!' SM17 yell-PRF SM17 plead help Seabelo
int: '"Fire!" has yelled to call for help Seabelo.'

Purpose clause adjuncts are consistent with secondary depictive predicates in this way.

2.1.3 Manner adverbs

Manner adverbs, another vP-level adjunct, also pattern alongside purpose clauses and secondary depictive predicates in their inability to precede the thematic subject in standard QI clauses.

(19) 'Nothing, nothing,' said Gollum softly.

(20) *'Nothing, nothing,' said softly Gollum.

(21) 'Titi o molato,' go bua kgosi kapela. (Collins 1997:2)
'Titi SM.3SG guilty' SM17 say chief quick
'"Titi is guilty," said the chief quickly.'

(22) *'Titi o molato,' go bua kapela kgosi. (Collins 1997:2)
'Titi SM.3SG guilty' SM17 say quick chief
int: "Titi is guilty," said quickly the chief.'

The story is the same with adverbial embedded clauses. In Setswana, the adverbial meaning can be expressed in QI clauses as a circumstantial embedded clause (Creissels 2023). These clauses have been referred to in the descriptive literature as a participial clause (Cole 1955), with the specific adverbial in the following example 'loudly' being expressed with a PP modifying the embedded clause's verb.

(23) 'Dumela,' go bua Seabelo a buela kwa godimo.
'hello' SM17 say Seabelo SM.3SG speak 17LOC above
'"Hello," says Seabelo loudly.' (*lit*: 'Hello,' says Seabelo, saying it loudly)

(24) *'Dumela,' go bua a buela kwa godimo Seabelo.
'hello' SM17 say SM.3SG speak 17LOC above Seabelo
int: '"Hello," says loudly Seabelo.'

Crucially, the adverbial circumstantial embedded clause must come after the AGENT in QI clauses. This is not the case for every single kind of embedded clause (see sections 3.2, section 4), but it is the case for those which are vP adjuncts. The same is true of English.

(25) a. 'Nothing, nothing,' said Gollum, muttering all about.
b. *'Nothing, nothing,' said muttering all about Gollum.

Thus far, the data support a unified analysis of argument structure and verb movement in both Setswana and English QI clauses. From all of these examples, it is clear that material that is not internal to (i.e., above) the VP does not move to precede the external argument AGENT in QI. I now solidify these observations by showing that VP-internal material reliably moves to precede the AGENT in QI clauses.

2.2 Control clauses

Control clauses, which are straightforwardly analyzed as VP complements, must appear before the AGENT in both English and Setswana QI clauses. This has previously been noted for English (Collins & Branigan 1997; Collins 2003). These facts are a marked difference from what we saw above with the embedded adverbial clauses.

(26) a. 'Help!' tried to yell Alex.
b. *'Help!' tried Alex to yell.

(27) 'Ke a go rata,' go batla go bua Seabelo.
'1SG DISJ 2SG like' SM17 want SM17 say Seabelo
'"I love you" wants to say Seabelo'

(28) *'Ke a go rata,' go batla Seabelo go bua.
'1SG DISJ 2SG like' SM17 want Seabelo SM17 say
int: '"I love you" wants Seabelo to say.'

Control clauses stand out from the other kinds of embedded clauses examined above because control clauses are complements – not adjuncts – of VP. As such, they are located in the projection that is moved into Spec,VoiceP for smuggling. The previous kinds of embedded clauses are located outside of the constituent that moves. Therefore, while the control clauses come to precede the in-situ AGENT, any embedded clauses located outside the VP remain following the AGENT.

2.3 Particles

The distribution of particles in English QI clauses provides additional support of the variety that control clauses give. As VP complements, particles in QI constructions cannot appear in a position following the AGENT (Collins & Branigan 1997; Collins 2003). Setswana does not have verbal particles in the same way that English does (or, at the very least, anything comparable does not exhibit the properties that are of interest here), so this section only pertains to English. The following examples are taken from Collins (2003:4).

(29) a. 'Where are you going?' called out John.
b. *'Where are you going?' called John out.

(30) a. 'He was wrong,' pointed out John.
b. *'He was wrong,' pointed John out.

(31) a. 'We can still win,' summed up the coach.
b. *'We can still win,' summed the coach up.

These facts are initially unexpected given the behavior of English particles, which may typically appear on either side of a postverbal thematic DP.

(32) John called (out) your name (out).

(33) John pointed (out) the problem (out).

(34) The coach summed (up) the meeting (up).

These data reveal a few interesting aspects of QI clauses, all of which clearly endorse an analysis of VP movement to a position above the AGENT. First, the distribution of particles in QI clauses suggests that we are dealing with a structure more complex than if the AGENT were externally merged low, and the THEME were externally merged above it: otherwise, we might expect the particles to pattern as they do in (32)-(34) for QI.

The position of particles in QI clauses additionally rules out a head movement analysis of verb raising in QI (Collins & Branigan 1997), as head movement for verb raising necessarily strands particles (Koopman 1991). Given that the particle is not stranded, it must be the case that the verb comes to precede the AGENT via phrasal movement (Collins 2003; Bruening 2016), as phrasal movement is what causes particles to move with their verbs (Van Riemsdijk 1978; Koopman & Szabolcsi 2000).

It is additionally worth noting that the distribution of particles in QI is characteristic of other inverted word order derivations which have been argued to involve VP phrasal movement, such as passives (Collins 2005) and dative shift (Collins 2021), as well as antipassives, middles, and locative inversion (Storment 2025). This fact will be crucial for the discussion of smuggling in section 4.

Lastly, the fact that a particle must precede the AGENT is evidence that the AGENT remains in-situ in QI (Collins 2003; Storment 2025), despite the fact that many analyses claim that the AGENT moves into the Infl domain in English QI (Collins & Branigan 1997; Bruening 2016), a question that I return to in detail in the following section. As I demonstrate in the following section, however, there is ample evidence to suggest that the AGENT remains in-situ in both Setswana and English, despite the facts on English verbal agreement.

3 In-situ status of the external argument AGENT

In the previous section, I have presented several data points showing that the VP in QI clauses in both English and Setswana moves to a position above the external argument AGENT. This analysis is consistent with other analyses of QI in English (Collins 2003; Bruening 2016). A major point of analyses of English QI clauses such as Bruening (2016), however, is that the postverbal AGENT in QI clauses must be located in Spec,TP. In this section, I show that that cannot be the case. Specifically, I empirically support the idea that the external argument AGENT in QI clauses remains in its base position (which I define as Spec,vP). Alternatively, I show that the quotative THEME is what moves to the structural subject (Spec,TP) position in QI.

I motivate these claims in terms of subject-verb agreement and Case assignment (3.1), the licensing of parasitic gaps (3.2), the raising of the THEME (not the AGENT) (3.3), and the Bantu conjoint/disjoint alternation (3.4).

3.1 Agreement

Perhaps the most striking difference between QI sentences in English and Setswana is found in the domain of verbal agreement. In Setswana, verbal agreement in QI clauses can never track the postverbal AGENT. In English, however, the prescriptive tendency (and easily just the tendency overall) is for verbal agreement to track the postverbal AGENT, though, as I show here, this agreement tendency is disrupted for many English speakers in a variety of circumstances.

3.1.1 Setswana agreement

In Setswana, verbal agreement, which is expressed via preverbal subject markers (SM) that track the noun class of the agreeing element (Halpert 2019), always surfaces as class 17[2] in QI clauses. SM17 has been analyzed as a default or expletive subject marker (Creissels 2011; 2017), and has been descriptively referred to as a lack of agreement altogether (a claim that I reject) given that it is also seen in infinitival constructions (Cole 1955). See the following contrast between non-inverted clauses and QI clauses.

(35) Dikgomo {di/ *go} bua, 'Mmuu.'
Cows {SM10/*SM17} say 'moo'
'Cows say, "Moo."'

(36) 'Mmuu,' {go/ *di} bua dikgomo.
'moo' {SM17/*SM10} say cows
'"Moo," says the cows.'

(37) (Nna) {ke/ *go} re, 'Dumela.'
(I) {SM.1SG/*SM17} say 'hello'
'I say, "Hello."'

(38) 'Dumela,' {go/ *ke} bua nna.
'hello' {SM17/*SM.1SG} say I
'"Hello," says I.'

For non-inverted clauses, the subject marker must obligatorily track the thematic subject. In QI clauses, however, the subject marker always surfaces as class 17, and can never track the features of the thematic subject. Given that Setswana subject markers straightforwardly index the element in Spec,TP (especially when that element is a standard referential DP) (Demuth & Mmusi 1997; Creissels 2011), the fact that agreement with the AGENT is not possible at all in QI clauses is direct evidence that the Agent has not moved into the structural subject position.

We can see that SM17 is true agreement with the quotative THEME by looking at another case in which the THEME is unambiguously functioning as the subject: a passive. In passives, it is known that the internal argument THEME functions as the structural subject See the following example of a quotative THEME passive subject, which also shows SM17 agreement.

(39) 'Go na le mathaithai kwa pele,' *go* tlhagisits-w -e ke Seabelo.
'SM17 be with puzzles 17LOC ahead' SM17 warn -PASS-FV by Seabelo
'"There is danger ahead," was warned by Seabelo.'

[2] In Setswana (and many related Southern Bantu languages), subject markers for noun classes 15 and 17 are said to be homophonous. Distinguishing the two cases actually proves to be not-very-straightforward and laden with theoretical assumptions that I consider somewhat peripheral to the arguments of this paper. For the sake of simplicity, I gloss all cases of this subject marker as noun class 17.

Given the observed similarities with the structure of English and Setswana QI clauses thus far, the fact that Setswana QI verbal agreement can never index the AGENT suggests that the situation with English QI—in which verbal agreement with the AGENT is very much possible—is not as straightforwardly analyzable as the AGENT being located in Spec,TP. I expand on this notion now.

3.1.2 English agreement

What we see in Setswana is the opposite of the prescriptive tendency and linguistic norm for agreement in English QI, in which verbal phi-agreement tracks the features of the AGENT. See the following example (including grammaticality judgments) from Collins & Branigan (1997).

(40) 'Avoid beef,' {advise/*advises} all the New Age dieticians.

Verbal agreement in Setswana QI can never track the thematic subject, and it has been purported that verbal agreement in English QI must always track the thematic subject. This difference seems to be at the core of claims that QI in English necessarily involves the AGENT moving into the Infl domain. This claim appears in Buell (2006), who claims such a derivational distinction between QI in Zulu, where verbal agreement always surfaces as default SM17 , and QI in French, where verbal agreement necessarily tracks the features of the thematic subject. This issue is additionally related to a long discussion in the literature on locative inversion in English and Bantu, where English can be contrasted with Bantu languages in which verbal agreement tracks the locative (and a further distinction can be made between Bantu languages in which the subject marker tracks the locative subject vs. Bantu languages where the locative inversion subject marker always surfaces as a default) (Bresnan & Kanerva 1989;Demuth 1990; Demuth & Mmusi 1997; Buell 2006; 2007; Zerbian 2006; Creissels 2011; Zeller 2013; Halpert 2015; 2019; Diercks 2017, a.o.).

Rather than claiming that these differences in agreement are indicative of completely different derivations of QI in Setswana and English, I maintain that these differences in agreement are actually superficial, and that the derivations of QI in English and Setswana are still underlyingly the same. Additionally, data from colloquial English suggest that agreement in English QI is more similar to Setswana than it would seem just by looking at the example and reported judgments in (40).

Storment (2025) provides convincing evidence that verbal agreement in English QI clauses need not necessarily track the features of the thematic AGENT. This evidence comes in the form of colloquial Internet data from native English speakers (native speaker status verified where possible) taken from social media, the grammaticality of which has been confirmed independently by native English speakers. For discussion on the validity and use of this kind of social media data, see Kilgarrif & Grefenstette (2003); Schütze (2011); Wikström (2017); Fernández-Serrano (2022); Collins (2023). See the following data, examples of which can easily be found *en masse*. Orthographic inconsistences from these naturally-occurring examples are preserved here.

(41) a. 'God bless you' says the men. (Database) (Storment 2025:139)
b. 'Go ahead! Spread that seed! Make more of me!' says the men's genes. (Blog)

c. ‘Where does it come from’ asks the children. (Museum piece)
d. ‘What for?’ asks the detectives. (Newspaper)
e. Quack Quack goes the Ducks, Moo Moo goes the Cows // Hop Hop goes the Rabbits and Cluck Cluck goes the Chickens (Pet food store)
f. ‘Thank you very much’ - says both organisms (Twitter)
g. ‘Mrs carter! these games were so fun thank you!’ says the kids who were absolutely behaving during the Halloween party in your class and not hyper at all (Etsy review)
h. It's my Grandfather's birthday!! ‘Happy birthday to you!’ sings the singers. (Blog)
i. ‘Virginity is made up’ says the people who think there are 200 genders. (Twitter)
j. ‘Time to get up’ says the boys. (Twitter)
k. ‘It can send a big clot running up to your lung…,’ warns doctors. (Blog)
l. Do you squeeze your spots? Doing so could KILL you, warns experts. (Blog)
m. ‘Many aspects of our psychology have evolved to be flexible rather than best suited to one specific way of life’, warns experts. (Blog)
n. Elderly motorists may be caught out by new DVSA eyesight proposals in a concern for those later in life, warns experts. (Blog)
o. Trump’s nuclear weapons strategy could be dangerous, warns experts. (Blog)
p. ‘Can you really do that and not kill your career’ asks parents everywhere. (News)
q. ‘Let me help you create a killer resume to beat the ATS’ - says people who sell resume writing services. (LinkedIn)
r. ‘Anyone who doesn’t share my exact opinion can’t be a member of the same broad political grouping as me’ says people who continue to import terms like TINO directly from other Parties in other countries (Reddit)
s. ‘Hey, I bet they’re out to get me!’ says people who are out to get others. (Blog)
t. ‘My5tery Music will bring out the Musical My5tery in you’ says three of our new step-team participants of Old West End Academy (Facebook)

It must be noted that not every English speaker accepts these examples, and they are certainly degraded in a formal register, although the judgments of the author (and other native speakers) are such that these are acceptable utterances. A lack of agreement here is also crucially optional: speakers who accept the agreement patterns in (41) also tend to accept plural agreement in the same configurations.

There seem to be several factors that causes the judgments around these examples to be a bit odd/contentious, including the fact that QI is associated with a “high” register, while a lack of phi-agreement with the thematic subject in English is typically associated with a “lower” register (e.g., ‘There’s problems’ instead of ‘There are problems’). In addition to this kind of register clash, there is also the issue of QI sentences being relatively infrequent and restricted to a narrative and/or literary context. Additionally, most English speakers are used to seeing QI clauses in the past tense (where phi-agreement is not visible), so examples of QI in the present tense with a non-third-singular AGENT are even more infrequent. All of these factors contribute to a potential lack of clarity around grammaticality judgments for the examples in (41). Despite the fact that many English speakers may not find them acceptable, I maintain that they are productive examples in many peoples’ English given the ease at which naturally occurring examples can be found online and the fact that a significant number of English speakers do find them acceptable (including the author), as evidenced by grammaticality judgment surveys, presentation of the data, and the availability of the data online.

Like in Setswana, in English, there is typically a straightforward relationship between subject-verb agreement and being located in Spec,TP, especially when dealing with a standard, non-phi-deficient referential DP (i.e., not something like a locative, predicate, or an existential expletive occupying the structural subject position). Thus, the fact that, for many speakers, agreement is optionally able to not track the features of the postverbal AGENT is a clear indicator that that AGENT is not what is located in Spec,TP.

The facts concerning Case assignment in English QI clauses, which I present now, further confirm this hypothesis.

3.1.3 English Case assignment

There is additionally a context for QI in which verbal agreement cannot track the AGENT, which, from what I can tell, is acceptable even for those people who would reject the examples (41). This context is retort-style sentences which make reference to a quotation that was previously established in the discourse (Lee-Goldman 2011).

(42) a. Says us![3] (Video) (Storment 2025:140)
b. Says them! (Video)
c. Says me! (Video)
d. Says you! (Video)

Even with nominative Case in such examples, which is perhaps marginally possible for some, it is not grammatical for verbal agreement to track the AGENT. See the following example with the author's judgments indicated.

(43) {Says/*say} {??we/??they/?I/you/y'all}!

It seems, in fact, that both nominative (44) and accusative (45) Case are available for postverbal pronominal AGENTs in English QI clauses, perhaps depending on the speaker. I include second person pronoun examples in (44), even though they are ambiguous in English between being nominative or accusative.

(44) a. 'I bet you can't spell my name,' says I. (Huckleberry Finn) (Storment 2025:141)
b. 'I bet you what you dare I can,' says he. (Huckleberry Finn)
c. 'All right,' says I, 'Go ahead.' (Huckleberry Finn)
d. 'To be sure we do' says we; for we never thought of playing such a dirty trick (The Catholic Layman)
e. 'How far you going?' says she with a toss of her rainbow mane. (Blog)
f. 'You never give us none' says they. (Newspaper)
g. 'There's a twenty pound reward' says you. (Besant 1902)
h. 'You're a fool' says you. (Besant 1902)
i. 'Let me only just find a door-mat,' says you to yourself. (Blog)
j. Oh, yes, says we to ourselves, Sydney's the place! (Blog)

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[3] One anonymous reviewer points out that that these examples here are obvious responses to the (rhetorical) question '*Says who?!*'. While that is obviously true, that is no reason that these configurations are not straightforwardly analyzable as quotative inversion, which I assume that they are.

k. ‘Now what do we do?’ asks I. (Blog)
l. ‘Should I be afraid of you?’ asks I over plastic bowls of salad. (Blog)
m. Where are you from? Asks I. (Twitter)
n. *Well*, asks I of myself, *what shall I find?* (The National Magazine)
o. ‘Why,’ thinks I, ‘what's the row? It's not a real leg, only a false leg.’ (Moby Dick)

(45) a. I wash my car too much, says me. (Facebook) (Storment 2025:140)
b. Hope I didn't lame it up too much, says me. (Blog)
c. Island gorls we up! (Says me, a person who has never been on an island) (Reddit)
d. Fire that sucker up, says me! (Reddit)
e. The industry could learn a thing or two about purpose-driven initiatives like these, says me. (LinkedIn)
f. ‘Well yes, but see if you can get a fee out of them’, says me. (Blog)
g. ‘I'm not gay’ - says me, a gay guy. (Reddit)
h. ‘I don't get it,’ says me. ‘How could Derrida die?’ (Blog)
i. ‘Probably too much’ says you! (Facebook)
j. ‘Giddy up’ says us. (Instagram)
k. Mail slots on the street? Mail and packages go there, says them. (Reddit)
l. ‘I’m up. I’m up’ says him, very OBVIOUSLY NOT UP. (Blog)
m. ‘Someone keeps moving the pot of gold, someone is not getting the memo, I’d rather be close to you.’ sings me to my India (Blog)
n. ‘How are you?’ asks me. (Blog)
o. ‘Pop!’ goes me (Song lyrics)

In fact, regardless of the Case features of pronominal AGENTs in QI clauses, it is possible to not have agreement track a pronominal AGENT – indeed, it may not be possible at all for first and second person pronominal AGENTs (46).

(46) a. *‘What do we do now?” ask {we/us}.
b. *‘I wonder what’s over there’ think {I/me} to myself.
c. *‘I feel great!’ scream you after a few drinks.

These examples represent a marked departure from what has been noted in the literature. Collins & Branigan (1997:7) note that nominative pronouns are preferred over accusative pronouns in English QI, and that first and second person pronouns are not grammatical at all, though the data suggests that is not the case for many English speakers.

What is clear from these examples is that the external argument AGENT is not located in Spec,TP, despite the fact that it can be agreed with in English. In English, there is a one-to-one correspondence between nominative Case assignment and being located in the structural subject (Spec,TP) position. Barring certain coordinate structures (see Caha 2024, a.o.), English nominative Case morphology is only seen on pronominal DPs located in Spec,TP, and pronominal DPs located in Spec,TP cannot appear with non-nominative Case morphology. If the external argument AGENT were simply located in Spec,TP in QI clauses, it would not make sense that accusative pronouns can so easily function as postverbal external arguments in these derivations. Furthermore, it would not make sense that nominative pronominal forms are of marginal/dubious grammaticality in these positions (as is corroborated by Collins & Branigan 1997:7).

The external argument AGENT in Setswana can never participate in subject-verb agreement, and the external argument AGENT in English can optionally not participate in subject-verb agreement and can easily not appear with nominative Case morphology. These facts clearly suggest that this AGENT never actually moves to the position of the structural subject. I motivate an analysis of agreement in QI that explains these facts.

3.1.4 Agreement analysis

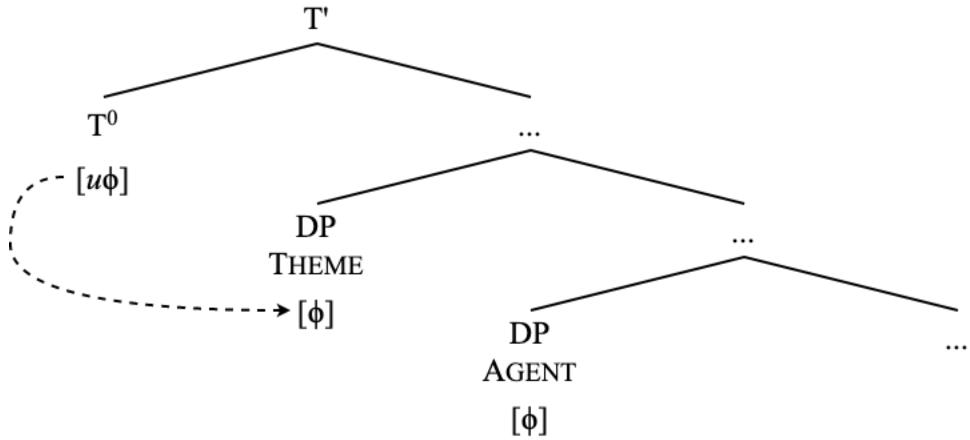
Given these novel data from Storment (2025), it becomes clear that verbal agreement in English QI constructions can indeed surface as a default value – third person singular – despite the phi-features of the Agent. In this way, English QI and Setswana QI are unified in that verbal agreement can surface as default value. The only difference is that it is obligatory in Setswana while it is optional in English (at least for third person non-pronominal DPs). See the following generalization.

(47) *Agreement in QI clauses*
In Setswana and English, verbal agreement in QI clauses may surface as a default value (SM17 or 3SG), either obligatorily (in the case of Setswana) or optionally (in the case of English).

The difference in obligatory default agreement in Setswana and conditionally optional default agreement in English QI clauses does not necessarily suggest that the structure of QI clauses in each language should be fundamentally different, especially given the structural similarities between these languages shown in section 2. Instead, I show that the differences in agreement can be reduced to a superficial difference in the nature of agreement probing in these two languages.

As such, verbal agreement probing starts out much the same in both languages, with probing being driven by a function head T^0 searching for an element bearing valued instances of [ϕ], as is standard to assume (Chomsky 2000; 2001; Storment 2025, a.m.o.). Given that the VP containing the quotative THEME has moved to a position above the external argument AGENT (as shown in section 2), it is crucial for this analysis of agreement that the quotative THEME be visible to the phi-probe on T^0 before the AGENT. Here, just for the discussion of Agree, I remain somewhat agnostic about the exact position of these elements with respect to one another, as all that matters for Agree—and all I mean to indicate in the following tree—is that the THEME be asymmetrically located above the AGENT at the time of probing by T^0 (despite that not being where the THEME is base-generated); I return to the finer details of this derivation in section 4.

(48) T^0 initiates a search for an element bearing [ϕ], encounters the THEME



Here, it is necessary to incorporate some notion of a defective/deficient goal that triggers default agreement. I adopt the following definition of a defective goal from Roberts (2010).

(49) *Defective goal* (Roberts 2010:62)
A goal G is defective iff G's formal features are a proper subset of those of G's probe P.

Let us say that the quotative THEME is phi-deficient, as evidenced by the fact that it triggers default agreement in both Setswana (always) and English (conditionally). The deficiency is additionally evidenced by the fact that coordinated quotative THEMEs cannot trigger plural agreement. The class 10 subject marker *di* is a kind of default plural subject marker in Setswana.

(50) *'I love you!' and 'Don't go!' yell the man.

(51) *'Dumela!' le 'Le kae?' di bua Seabelo.
'hello!' and 'and where?' SM10 say Seabelo
int: '"Hello!" and "How are you?" say Seabelo.'

In fact, no noun class prefix of any kind can appear on the quotative THEME in QI, further suggesting its defectiveness.

(52) *Di 'Dumela!' di bua Seabelo.
CL10 'hello!' SM10 say Seabelo
int: 'The "Hello!"s say Seabelo.' (i.e., Seabelo says 'Hello!' multiple times)

Similarly, quotes cannot be marked as plural in English[4].

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[4] There is a peripheral phenomenon in English in which something like looks like a quote can appear with number inflection, and also with a determiner and even adjectival modification.

(i) I remember fondly all the warm "I love you"s that my husband used to say to me.

(53) *'I love you's says my husband every day.
(*int*: My husband says 'I love you' multiple times per day every day)

As it is relevant for English, the quotative THEME lacks a number feature. As it is relevant for Setswana, the quotative THEME lacks a number and a gender (≈noun class) feature.

Regarding the phi-probe on T^0, we can say that this probe in Setswana is "satisfied" (e.g., Deal 2024) by any goal bearing any instance of phi at all, but the lack of any number and gender features internal to [ϕ] will trigger the default subject marker *go* at spellout. We see this subject marker additionally occur with expletive subjects (54), inverted locative subjects (55), a stipulated large PRO of infinitival clauses (56), and sentential subjects (57).

(54) *Go* a na.
SM17 DISJ rain
'It's raining.'

(55) Kwa maemelong a diterena *ga* goroga terena.
17LOC station of trains SM17.PST arrive train
'At the station arrived a train'

(56) Ke batla [PRO *go* ithuta Setswana]. (Cole 1955:309)
SM.1SG want [PRO SM17 learn Setswana]
'I want to learn Setswana'

(57) [Go roga batho jalo] *go* tla go tshwar-isa mapodise. (Cole 1955:438)
[SM17 curse people thus] SM17 FUT you arrest -CAUS police.PL
'Cursing people like that will cause the police to arrest you'

Cole's descriptive grammar of Setswana even notes that the "infinitive" noun class is "non-indicative of number" (Cole 1955:68), further corroborating the phi-deficiency of subjects which trigger *go* as the subject marker. In fact, Creissels (2017) specifically adopts an analysis that the *go* morpheme in (56)—and in the embedded clause in (57)—is straightforwardly a default SM17/SM15 subject marker, as opposed to some kind of homophonous infinitival morpheme, which is the analysis that I implicitly adopt here ass well.

The examples in (54)-(57) all involve a nonstandard element standing in as the structural subject: an expletive, a locative, PRO, or a VP. These elements—along with quotative THEMEs—are consistently not nominals, or else some kind of deficiently represented nominal; elements that are not canonical subject DPs. Thus, the stipulation of these elements as phi-deficient is not arbitrary and can instead be tied to some structural commonality, potentially such as the lack of a [D] feature and thereby the lack of a DP projection (though, at this point, this specific structural deficiency is purely stipulative; tests for diagnosing D-less nominals would have to be employed here (i.e., Pereltsvaig 2006; Neumann 2025)).

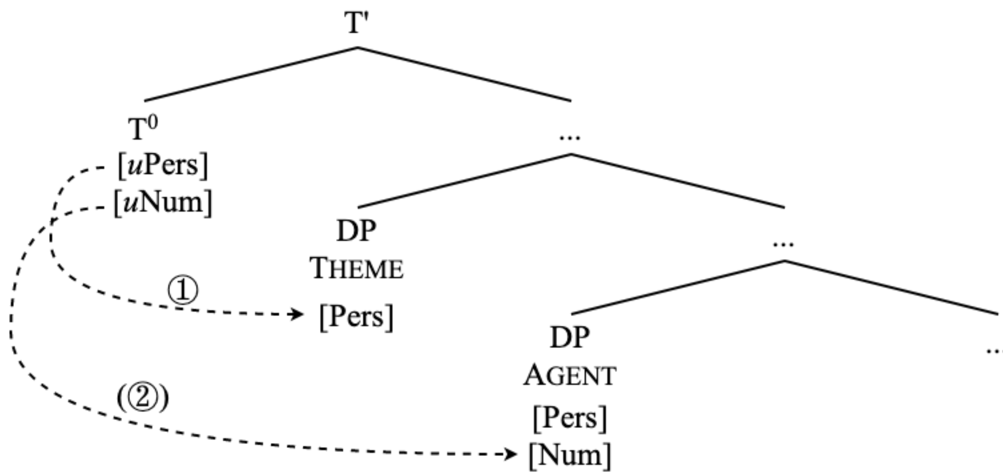
These nominalized quotes seem to only work with short, oft-repeated phrases, and they definitely cannot participate in QI as structural subjects. I assume that such lexicalized quotations are a phenomenon not directly related to quotes as I discuss them here pertaining to inversion, though they certainly are interesting.

Returning to the schema in (48), then, T^0 in Setswana having probed and undergone Agree for [ϕ] with the quotative THEME, the probe is valued for [ϕ] and is thus fully “satisfied”, and the lack of number and gender features internal to phi triggers deficient agreement *go* at spellout.

As for English, the T^0 probe is relativized to be probe not just for [ϕ], but for person and number features (Béjar 2003; Béjar & Rezac 2009; Preminger 2014; Coon & Keine 2021; Storment 2025, a.o.). As the quotative THEME lacks a number feature (but can be reasonably assumed to have a third person feature, as third person entails only being neither a speaker nor an addressee, which is certainly true of a quote), then the search by T^0 that finds the quotative THEME in (48) will value person, but not number.

The probe could stop searching here, thus triggering third person singular at spellout (third person from the quotative THEME and a default number value), or it could cyclically probe again for number. This is schematized below.

(58) Conditional cyclic probing in English QI.



If this second cycle of probing occurs, then agreement may track the features of the AGENT as well as long as the features of the AGENT are compatible with the features of the quotative THEME that the probe has already acquired from the first probing cycle.

As such, a second probing cycle agreeing with a third person plural AGENT copies back features to the probe that are a superset of the features acquired from the THEME ([3, pl] ⊃ [3]). Given this compatibility of features, the cyclic probing of a third person plural AGENT results in the verbal agreement in a QI clause being spelled out as third person plural.

This process is known as defective circumvention (Storment 2025), by which a probe can conditionally search again after having agreed with a defective goal and go on to agree with a further, non-defective goal. This is formalized below.

(59) *Defective Circumvention* (Storment 2025)
A probe P enters into an Agree relation with a higher phi-deficient goal α and then conditionally goes on to Agree past α with a lower, more featurally-specified goal β.
1. Agree without defective circumvention → 3sg “default” agreement
↳‘“What?!” asks the men.’
2. Agree with defective circumvention → agreement with postverbal AGENT
↳‘“What?!” ask the men.’

In this system of conditional cyclic probing, defective circumvention can fail to obtain when the features of the lower goal are incompatible with the features of the higher defective goal. For example, if the AGENT has a first or second person feature, then that would be incompatible with the third person feature already acquired from the quotative THEME, as [1/2, sg/pl] $\not\supseteq$ [3].

As such, the second probing cycle would produce an irresolvable and uninterpretable (and un-spell-out-able) feature conflict on the T^0 probe. Defective circumvention with a first or second person AGENT—as in (46)—in QI clauses, then, is ruled out for these reasons, and the only viable derivation is one in which the probe does not search a second time (thus leading to default agreement). This restriction essentially derives weak PCC effects with multiple subject agreement in QI clauses (Storment 2025). For a descriptively similar proposal of PCC effects driven by multiple Agree, see Coon & Keine (2021). For a more detailed discussion of how exactly the second cycle of probing could stand to be optional in the grammar, see chapter 2 of Storment (2025).

As it is relevant for our discussion here, in this system, there is no need for the lower goal to actually have moved into the Infl domain at any point (and, indeed, this movement would be blocked by prior movement of the *first* element to have been probed by T^0). In fact, the fact that first and second person DPs cannot be agreed with at all in QI English clauses—see (43), (46)—strongly suggests on its own that the AGENT does not move into the Infl domain, as there is a one-to-one correspondence between a DP being located in Spec,TP and a DP controlling verbal phi agreement in English. See the related case of existentials.

(60) a. You {were/*was} there.
b. There {was/*were} you.

Furthermore, the fact that AGENTS do not obligatorily surface with nominative Case in English QI clauses is evidence that they do not move into the Infl domain – see (42), (45) – as there is an additionally clear association in English between Spec,TP and nominative Case forms.

(61) a. {I/*me} was there.
b. There was {me/??/*I}.

Given that phenomena such as verbal phi-agreement and Case assignment are what we call A-dependencies (Fong & Halpert 2023), meaning that they are associated with A-movement, the default agreement in Setswana and English is reliably taken to be an indication that the default-agreement-triggering element – the quotative THEME – has undergone A-movement to the position of the QI clause's structural subject, with the AGENT remaining in its base position. The fact that AGENTs in English QI clauses do not reliably appear with nominative Case is evidence that those AGENTs have *not* undergone A-movement to Spec,TP.

The availability of accusative Case for AGENTs in English QI is straightforwardly explained by the fact that they do not move into Spec,TP. As for the fact that nominative Case is somewhat available for these AGENTs, I believe that this can be explained as a kind of high-register "default" nominative (Caha 2024) which does not actually require movement into Spec,TP, and this analysis is evidenced by the alternation with accusative forms.

Unsurprisingly, the AGENT in Setswana QI clauses remains in-situ as well, thus mirroring analyses of locative inversion in Setswana and related languages (Creissels 2011, a.o.).

If the Agent does not appear in the structural subject position, and the THEME is first probed for subject-verb agreement, it stands to reason that the THEME moves into this position. I support this view in the following subsections. The A-movement of the THEME to Spec,TP is supported by the failure of the quote to license parasitic gaps in English (3.2), and the data on raising in English and Setswana (3.3). The fact that the AGENT remains in situ is further evidenced by the data surrounding the mysterious conjoint/disjoint alternation in Bantu (3.4).

3.2 Failure to license parasitic gaps

A concise piece of evidence from English that the quotative THEME undergoes A-movement is the fact that the preposing of the quote in QI clauses fails to license a parasitic gap, which is not true of standard quotative preposing in a non-inverted clause. Ā-movement is the only movement that licenses parasitic gaps, not A-movement (Murphy 2022).

(62) a. *'We should leave,' thought Max without actually saying __. (Murphy 2022:4)
b. 'We should leave,' Max thought without actually saying __.

Compare (61a) to (63), where a resumptive pronoun fills the gap.

(63) 'We should leave,' thought Max without actually saying it. (Murphy 2022:4)

As such, the quotative THEME in QI undergoes A-movement to its preverbal position, while the fronted quote in non-inverted clauses does not. Further evidence of this claim is that, in non-QI clauses with a fronted quote, there is no possibility of having verbal agreement not track the AGENT. Recall example (7), modified here to show the relevant agreement facts.

(64) 'Sméagol has to take what's given to him,' the Hobbits {answer/*answers}.

Indeeed, the same is true of Setswana non-QI clauses with a preposed quote.

(65)	'Le kae?'	Seabelo	{a/ *ga}	botsa.
	'and where?'	Seabelo	{SM.3SG.PST/*SM17.PST}	ask
	'"How are you?" Seabelo asked.'			

A preverbal quotative THEME that fails to license parasitic gaps is one that can be agreed with. A preposed quote that licenses parasitic gaps cannot be agreed with. This is because the preverbal quote that cannot license parasitic gaps appears preverbally due to A-movement, which is associated with phi-agreement. The quote that licenses parasitic gaps has undergone Ā-movement, which cannot control verbal phi-agreement.

3.3 Raising

The next piece of evidence that I present here for the A-movement of the quotative THEME (and the in-situ status of the AGENT) is with subject-to-subject raising predicates. While there is a precedent in the literature for subject-to-subject raising diagnosing the A-properties of the preposed locative in locative inversion, a related construction (see section 6), in both English (Levine 1989; Culicover 1991; Coopmans 1992; Bresnan 1994; Diercks 2017, a.o.) and Setswana (Demuth & Mmusi 1997; Creissels 2011), subject raising has not received the same amount of attention in the concise literature on QI constructions. In this section, I discuss the possibility of subject-to-subject raising of the preverbal quotative THEME of QI clauses in English (3.3.1) and Setswana (3.3.2), showing that it is productive in both languages. The availability of subject raising is a clear A-property of the quotative THEME in QI clauses.

3.3.1 English

As mentioned, subject raising has been described in English locative inversion clauses, taken to be evidence that the locatives can undergo A-movement as structural subjects (Bresnan 1994).

(66) a. Over my window seems to have crawled an entire army of ants. (Bresnan 1994:96)
b. On that hill appears to be located a cathedral
c. In these villages are likely to be found the best examples of this cuisine

Despite the similar properties of locative inversion and QI, the issue of subject raising has barely been commented on in the literature on quotative inversion in English. In fact, the only mention of it outside of Storment (2025) that I could find is a footnote in Bruening (2016), who states that "one reviewer finds auxiliaries and even raising and control verbs fine with quotative inversion" (Bruening 2016:fn. 12). The implication in the text seems to clearly be that raising should not be predicted to be acceptable for QI clauses in Bruening's theory, though such examples are not actually discussed in the paper. As such, this can be taken as evidence that at least one person actually accepts raising with QI clauses. The judgments of the anonymous reviewer of Bruening's paper align with my own judgments, and, indeed, naturally occurring grammatical examples of subject raising in QI are readily available online.

(67) a. 'Where did the wheat go?' seemed to say the heron. (Flickr) (Storment 2025:138)
b. 'He…hello…' seemed to say someone behind the door. (Wattpad)
c. 'We shall live and see', seemed to say Lenin in the Great Initiative or in the State and Revolution. (Revista Cogito)
d. 'Who are you, what do you want,' seems to ask this little guy feeding under my bird feeder. (Forum)
e. 'Sorry', seemed to say the elephant with the cry elephants usually do. (Forum)

Thus, in English, a quotative THEME is compatible with subject-to-subject raising in QI clauses. Though naturally-occurring examples online with a raising predicate other than 'seem' are harder to come by, grammatical examples are still easily producible[5].

[5] It has been suggested to me that, in QI subject-raising examples in English, agreement between the matrix verb and the postverbal AGENT is more difficult to obtain than it is with monocausal QI examples.

(ii) a. 'Shut up!' says the rowdy schoolchildren.
b. 'Shut up!' say the rowdy schoolchildren.

(iii) a. 'Shut up!' is likely to say the rowdy schoolchildren.
b.?/*'Shut up!' are likely to say the rowdy schoolchildren.

(68) a. ‘What about the files?’ appeared to ask a member of the crowd.
b. ‘What the hell?’ is likely to say anyone who reads my diary.

Notably, such raising is not possible in non-QI clauses with a preposed quote (that is, where the AGENT precedes the verb.

(69) a. *‘Where did the wheat go?’ seemed the heron to say.
b. *‘What about the files?’ appeared a member of the crowd to ask.
c. *‘What the hell?’ is likely anyone to say.

The unacceptability of the examples in (69), contrasted with the acceptability of the (67) and (68) sentences, shows that subject raising is incompatible with Ā-movement of a quote and is thus only applicable to those quotative THEMEs which undergo A-movement to the structural subject position of the embedded clause of raising predicates.

3.3.2 Setswana

Setswana shows multiple subject agreement with certain aspectual auxiliary verbs (Cole 1955:236, 448; Lee & Hlungwani 2015:133). These aspectual auxiliary constructions are interesting in that they cause agreement morphemes to appear in multiple positions.

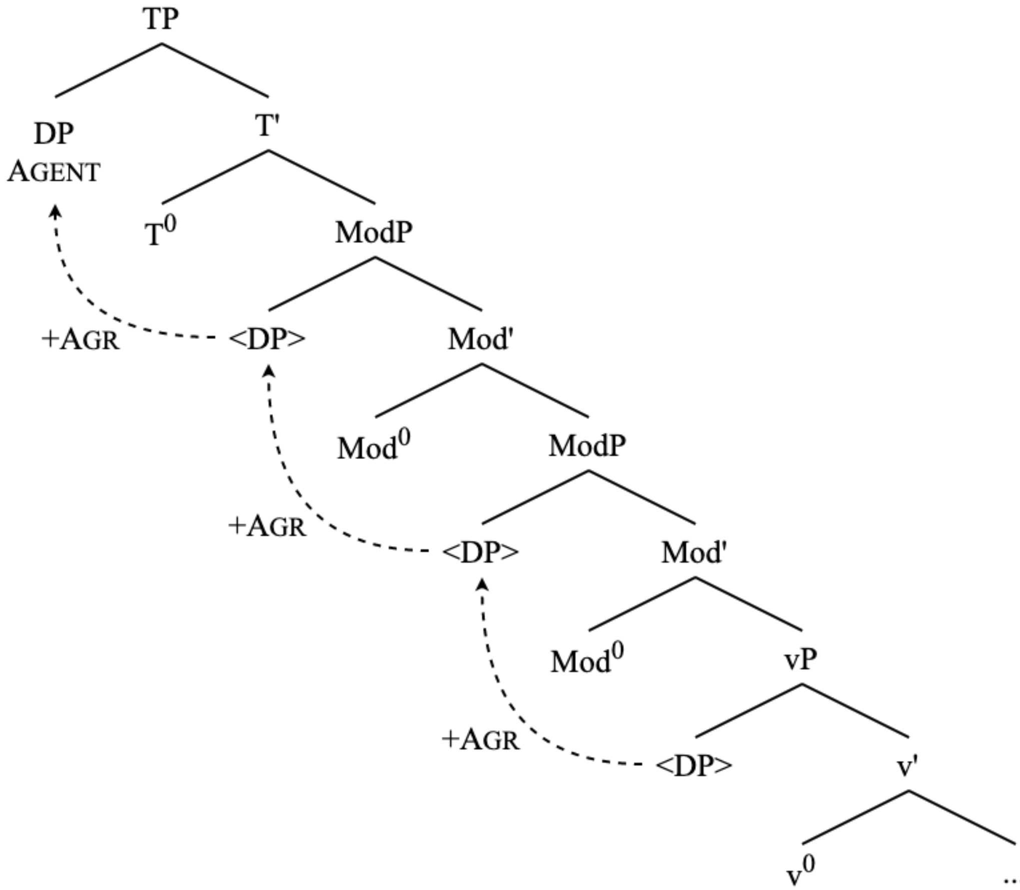
(70)	Re	tlaa.bo	re	santse	re	bala,	re	kwala	re	ithuta. (Cole 1955:448)
	SM.1PL	shall	SM.1PL	still	SM.1PL	read	SM.1PL	write	SM.1PL	learn

‘We shall still be reading, writing, and studying.’

While these constructions may not constitute “raising” in the narrowest traditional sense (i.e., each verbal “layer” may not have a full clausal structure associated with it), the phenomenon allows us to see multiple instances of agreement marking of the structural subject. Such multiple agreement has been taken to be evidence of clause-internal successive cyclic subject movement in which agreement is marked on all Tense/Aspect/Modal functional heads (Pietraszko 2024; 2025), which is claimed to be driven by EPP/Phi probes present on the relevant heads (Carstens 2001). Thus, following Pietraszko (2025), we can assign the following structure to (70).

I delegate these examples to a footnote because my own judgments on them are not so clear. One anonymous reviewer points out that their judgments sharply do not align with my own here, so I leave a finer discussion of these data for a later time.

(71) Successive cyclic raising with multiple aspectual adverbials



In this structure, each successive move of the subject is an instance of A-movement as it is driven by a search for Phi (van Urk 2015), which is indeed characteristic of all EPP movement in Bantu (Baker 2003; 2008; Carstens 2005; Pietraszko 2020; 2025). As such, the clause-internal raising is featurally motivated in the same way that raising out of a non-finite clause is. I choose, then, to focus on these examples of clause-internal aspectual auxiliary over classic raising examples as classic multiclausal raising in Setswana does not allow us to see instances of multiple subject agreement. Note the lack of subject agreement on the embedded verb in cases of traditional subject raising (*go* in the following example should be taken as an infinitival marker, and not an instance of defective agreement).

(72) Basadi$_i$ ba bonagala t_i go tsofala ka bonako. (Demuth & Mmusi 1997:6)
women$_i$ SM2 seem t_i SM17 age PREP fast
'Women seem to age fast.'

Turning to verbs of quotation, an AGENT that raises to the position of the clausal subject with a verb of quotation triggers successive cyclic agreement on any aspectual auxiliaries that are present in the clause, as expected.

(73) (Nna) ke nna ke re, 'Dumela.'
(I) SM.1SG stay SM.1SG say 'hello'
'I habitually say, "Hello."'

(74) (Wena) o nna o re, 'Dumela.'
(You) SM.2SG stay SM.2SG say 'hello'
'You habitually say, "Hello."'

(75) Dikgomo di nna di re, 'Mmuu.'
cows SM10 stay SM10 say 'moo'
'Cows habitually say, "Moo."'

(76) Seabelo o tswelela a re, 'Dumela.'
Seabelo SM.3SG proceed SM.3SG say 'hello'
'Seabelo continually says, "Hello."'

With inversion, however, agreement with the AGENT DP does not appear anywhere in the clause.

(77) 'Dumela,' go nna go bua nna.
'hello' SM17 stay SM17 say I
'"Hello," habitually says I.'

(78) 'Dumela,' go nna go bua wena.
'hello' SM17 stay SM17 say you
'"Hello," habitually says you.'

(79) 'Mmu,' go nna go re digkomo.
'moo' SM17 stay SM17 say cows
'"Moo," habitually says the cows.'

(80) 'Dumela,' go tswelela go bua Seabelo.
'hello' SM17 proceed SM17 say Seabelo
'"Hello," continually says Seabelo.'

Pietraszko (2025) observes the same property of Ndebele aspectual auxiliary constructions, noting that clause-initial AGENTs trigger agreement on all auxiliary elements.

(81) Ubaba u- be e- se e- si- dla. (Ndebele; Pietraszko 2025:9)
father SM1-AUX.PST SM1-AUX.PERF SM1-PROG-eat
'Father had already been eating.'

Additionally, clause-final AGENTs (with no preverbal subject material) cannot trigger agreement, with the default SM17 appearing on the auxiliaries instead.

(82) Ku- be ku- se ku- si- dla ubaba. (Ndebele; Pietraszko 2025:10)
SM17-AUX.PST SM17-AUX.PERF SM17-PROG-eat father
'Father had already been eating.'

Indeed, Setswana functions in the same way. No overt preverbal subject is required to obtain a string of SM17 marking on the auxiliaries.

(83) a. Seabelo o setse a tsene.
Seabelo SM.3SG already SM.3SG walk
b. Go setse go tsene Seabelo.
SM17 already SM17 walk Seabelo
'Seabelo has already walked.'

While Pietraszko (2025) takes forms such as (82) and (83b) to be a lack of agreement from nothing having undergone clause-internal raising at all, the presence of default subject marking with a postverbal AGENT in these constructions in Setswana and Ndebele can instead be taken as an instance of the relevant heads having undergone agreement with a defective goal – an expletive, in the cases of (82) and (83b), and the quotative THEME in (77)-(80). Assuming that there is an active EPP requirement on these functional heads (Baker 2008; Collins 2004; Pietraszko 2023; 2025), it follows that some element should move through these heads as an agreement goal, even when default agreement appears. Comparing examples like (54) to what we see here, it is clear that some deficient element should be undergoing clause-internal raising in the examples where default subject marking appears on aspectual auxiliaries with a postverbal AGENT, regardless of whether or not something appears in the preverbal position.

This discussion mirrors discussions of classic raising with locative inversion in Setswana (Demuth & Mmusi 1997; Creissels 2011), in which default agreement in the matrix clause is taken to either be agreement with the fronted locative PP (Demuth & Mmusi 1997) or with a null expletive subject (Creissels 2011). Either way, though, the appearance of SM17 is taken to be an instance of agreement with a deficient element, as opposed to a lack of agreement altogether. I return to the issue of locative inversion agreement in Setswana in section 6.

Agreement in QI clauses with aspectual auxiliaries (when contrasted with agreement in non-inverted quotative sentences) is a clear indicator of clause-internal successive cyclic A-movement (i.e., raising) of the deficient quotative THEME. In this way, quotative THEMEs in both English and Setswana are able to undergo raising, further cementing their status as A-moved elements, which follows from the account of QI as smuggling. Additionally, the *lack* of agreement with the AGENT in these Setswana subject raising examples further cements the analysis of the AGENTs remaining in-situ in QI clauses.

3.4 The conjoint/disjoint alternation

The final piece of evidence that I introduce for the AGENT remaining in-situ in Setswana QI clauses – and, by extension, the quotative THEME having undergone A-movement to the structural subject position – comes from data surrounding the mysterious conjoint/disjoint alteration in Bantu languages (Meeusen 1959; van der Spuy 1993; Creissels 1996; Ndayiragije 1999; Buell 2006; van der Wal 2017; van der Wal & Hyman 2017; Halpert 2012; 2017, a.m.o.).

Also descriptively known as the short/long form alternation (Cole 1955:243) (among many other names), the conjoint/disjoint alternation refers to the absence/presence of verbal morphology that appears under certain circumstances, including when the verb is the final element in a clause. The conjoint form is denoted by the lack of this morpheme (84), while the disjoint form is characterized by the presence of the morpheme (85).

(84) Dikgomo di fula kwa koneng. (Creissels 1996:109)
cows SM10 graze 17LOC river
'The cows graze at the river.'

(85) Dikgomo di *a* fula. (Creissels 1996:109)
cows SM10 DISJ graze
'The cows graze.'

I adopt the following definition of the alternation, from van der Wal (2017).

(86) *The conjoint/disjoint alternation* (van der Wal 2017:55)
The conjoint/disjoint alternation is an alternation between verb forms that are formally distinguishable, that are associated with an information-structural difference in the interpretation of verb and/or following element and of which one form is not allowed in sentence-final position.

Descriptively, the alternation can be characterized in this way: the disjoint form appears when the verb is in sentence-final position, and the conjoint form appears elsewhere. Analytically, the alternation is more difficult to characterize. In the literature, this alternation has been tied to Case licensing (Halpert 2012), information structure (Creissels 1996; Buell 2006), and prosody (Halpert 2017; Kusmer 2019). Throughout these different accounts, the generalization remains clear that the conjoint form surfaces whenever some material that is externally Merged within the vP – arguments and adjuncts alike – remains inside the vP. The disjoint form, on the other hand, surfaces whenever all vP-internal material has moved outside of the vP, for reasons related to licensing (A-movement) or information structure (Ā-movement) (Halpert 2012; van der Wal 2017).

As it is relevant to this discussion, the disjoint morpheme cannot appear in QI clauses.

(87) 'Dumela,' go (**a*) bua Seabelo.
'hello' SM17 (*DISJ) say Seabelo
'"Hello," says Seabelo.'

For the purposes of QI, the descriptive generalization[6] on the distribution of the disjoint morpheme is enough to demonstrate that the AGENT remains in its base position internal to vP in QI derivations in Setswana. If the AGENT moved out of its base position, then the vP would be empty, and the disjoint morpheme would appear, but it does not.

[6] I do not propose a syntactic analysis of the alternation here, as I do not have anything revolutionary to say about it.

3.5 Category of the quote

While we have thus far identified the A-movement of the quotative THEME to Spec,TP and the in-situ status of the AGENT, it is clear that quotes do not behave as normal subject constituents. There is no requirement that the quote itself actually appear in a preverbal position in QI clauses (Collins & Branigan 1997), which is surprising given the EPP requirement of the structural subject positions in English and Setswana (Carstens 2005).

(88) Said the night wind to the little lamb, 'Do you hear what I hear?'

(89) Ga botsa Seabelo, 'Le kae?'
SM17.PST ask Seabelo 'and where?'
'Asked Seabelo, "How are you?"'

Note that the V-AGENT word order is still retained in these examples.

Additionally, two parts of a single quotation may appear on either side of a QI clause, which is not something that can be done with a typical subject constituent.

(90) 'But,' said Frodo, 'That is a long way off.'

(91) 'Ngwana-ka,' go bua Moremi, 'Mma ke go gakolol-e.'(Cole 1955:451)
'child -my' SM17 say Moremi 'allow SM.1SG you advise -SBJ'
'"My child," says Moremi, "Let me advise you.'

Indeed, the boundaries of the quotation as it appears on either side of the verb and external argument do not even need to coincide with a constituent boundary inside the quote.

(92) 'I think that you…' said Mary, '…should go to the store tomorrow.'

QI stands out from other inversion constructions in this respect. See locative inversion in English as an example of this.

(93) a. *Walked John into the room.
b. *Into walked John the room.

Given these facts, it seems unlikely that the quote itself is actually occupying the position of the structural subject in QI clauses, as it behaves nothing like a standard structural subject. Instead, I adopt the analysis of Collins & Branigan (1997) that the structural subject in QI clauses—the element which starts out as an internal argument THEME—is a null quotative operator, a type of null argument that links the position of the thematic argument to a quotation that appears adjacent to the relevant clause, or elsewhere in the discourse (see the *Says who?!*-style examples in (42)).

Regarding the category of the quotative operator, it is straightforwardly a nominal element (Collins & Branigan 1997). While it may be tempting to claim that quotative verbal complements should be CPs in analogy with *indirect* speech reports (see Staps & Rooryck 2023), it is not the case that a quote should have the internal structure of a CP at all.

(94) ‘Beans!’ said John.

(95) ‘Sengwe!’ go bua John.
‘something!’ SM17 say John
‘“Something!” says John.’

In fact, a quote arguably does not need any complex internal grammatical structure at all.

(96) a. ‘Pop!’ goes the weasel.
b. ‘Hmmph,’ went Alex.
c. ‘Ahhhhh,’ cried John.

Quotes additionally do not even need to be grammatical, or conform to the grammar of the language of the main clause.

(97) a. ‘Who did a picture of hang on the wall?’ said the professor of linguistics.
b. ‘*Mis queridos estudiantes*,’ said the Spanish teacher, ‘*Hoy vais a tomar un examen*.’

Furthermore, indexical elements such as pronominals and deictics in indirect speech report clauses interact with the matrix clause in a way that they certainly do not with quotes.

(98) a. Alex$_i$ said (that) he$_{i/j}$ loves me$_{SPKR}$.
b. Alex$_i$ said, ‘He$_{j/*i}$ loves me$_{i/*SPKR}$.’

The ways in which direct speech reports and indirect speech complements differ from one another reveal two things. First, a quote is not part of the clause it appears with in the same way that a typical verbal complement is, which follows from the analysis the quote itself is not actually appearing in Spec,TP in QI clauses. Second, if the structure of clauses containing quotes is to be analogous with clauses containing indirect speech reports, then these observations should be taken as evidence that sentences in which an indirect speech report appears as a verbal complement also contains a covert nominal argument. The facts concerning clausal prolepsis (Angelopoulos 2025) support this view (99), as does the fact that an indirect speech report may cooccur with other kinds of nominal objects (100).

(99) It has been said that dragon-fire could melt and consume the Rings of Power.

(100) a. Mary said something shocking: that she loved me.
b. He finally said it: that he loves me.

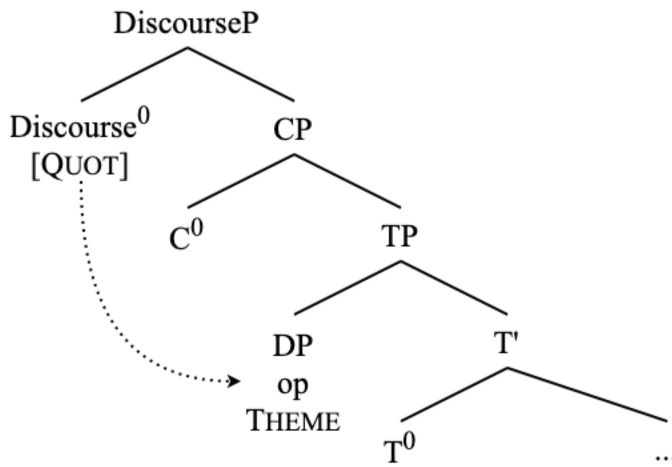
Sentences with quotative arguments behave much the same way.

(101) It has been said, ‘Only dragon-fire can melt and consume the Rings of Power.’

(102) a. Mary said something shocking: ‘I love you.’
b. He finally said it: ‘I love you.’

As such, standard QI clauses involve the movement of the quotative operator from the THEME position to the structural subject position after the VP has moved to a position above the AGENT. The actual quote, given that it does not appear in the constituent internal to the derivation itself (and need not actually be pronounced as part of the utterance at all), is not represented in the syntactic derivation of the utterance. The relationship between the quote and the operator—namely, that the operator should be bound by a quote that appears somewhere in the immediately relevant discourse (Collins & Branigan 1997; see also Suñer 2000)—is syntactically represented by a special discourse head bearing a [QUOT] feature to bind the quotative operator, assuming that discourse-related and pragmatic meaning should ideally be computed from an underlying syntactic structure (see chapter 10 of Collins 2024). Here I show how this looks specifically when the quotative operator THEME has moved to the position of the structural subject and how it is bound by the discourse quote feature.

(103) The status of the subject in QI clauses.



The examples in (101) and (102) as it pertains to quotative clauses in general, as well as the fact that it is possible in English to have preverbal particles such as *so* or *thus* in QI clauses (e.g., *Thus says the Lord your God*) suggest that the quotative operator is able to alternate with overt constituents in certain registers.

The licensing of a null argument—the quotative operator—in the subject position here is worthy of discussion as it pertains to English, a language which famously does not permit null structural subjects in most cases (this is less of an issue for Setswana, which allows pro-drop). If this were the only situation in the entirety of English where null-subjects are purportedly permitted, then this proposal may be somewhat problematic in this respect. Fortunately, however, that is not the case. The overt-EPP requirement of English is seemingly suspended in a variety of circumstances, including imperatives (104) (Downing 1969; Bekeuma & Coopmans 1989) and so-called 'diary-drop' (105) (Haegeman & Ihsane 1999; Haegeman 2013), certain prototypical retorts/replies (106), as well as informal registers of English more generally (Weir 2008).

(104) Come over here!

(105) Had that dream again last night.

(106) a. Love you too.
b. Wouldn't be the first time.

In all the cases shown here, the ability to not have an overtly pronounced structural subject is connected to the discourse functions of the clause; i.e., that the understood subject is obvious from the established discourse, and that the clause serves a particular purpose in the discourse such as expressing a command or augmenting an utterance in the discourse that immediately precedes it from another interlocutor. Indeed, these discourse functions have been implemented as explanations of the null subjects in these aforementioned cases (Bekeuma & Coopmans 1989; Haegeman 2013, a.o.).

The analysis of the quotative THEME in QI clauses as a null quotative operator (Collins & Branigan 1997) fits into the view of QI as smuggling that I motivate in the following section, and it allows for smuggling of an element other than the quote itself, which, as I have shown here, does not behave as if it were any kind of clause-internal constituent.

The null quotative operator also informs our understanding of the phi-deficiency of the quotative THEME as discussed in section 3.1. As a phonologically null operator that alternates with light nouns (see (102)), it makes sense that this element would be structurally deficient when compared to other nominal arguments, and the phi-deficiency follows from this structural deficiency. In fact, the operator is structurally similar to an expletive argument in this way, and in the way that is serves as a kind of placeholder argument for something else (the quote). I return to this parallel between expletives and the quotative operator in section 6, after a discussion in the following section of the transitivity constraint in QI clauses.

4 QI as smuggling

QI (107) is characterized by the presence of an internal argument in a preverbal subject position (which is an A-position, as I have defended in section 3), while the external argument of a clause surfaces postverbally, deriving a marked OVS word order—in which the external argument appears postverbally and the internal argument appears preverbally—which is a clear departure from the canonical SVO word order of English. The same pattern characterizes many other clause types as well: locative inversion (108), predicate inversion (109), and passives (110).

(107) 'Halt!' yelled Uglúk.

(108) Down from the wall leapt Gimli with a fierce cry.

(109) The first thing he saw was Gollum.

(110) Boromir was slain by the same orcs whom you destroyed.

These examples and their underlying derivations present issues for relativized minimality (Rizzi 1990). How does the internal argument THEME in QI come to precede the external argument AGENT via A-movement, assuming that internal argument THEMEs are uniformly projected in a position below external argument AGENTs?

The issues of minimality in QI and these similar derivations are ultimately unavoidable within a Minimalist theoretical framework. In this section, I demonstrate that smuggling presents a clear solution to them, in addition to naturally accounting for the empirical facts regarding VP movement and the A-movement of the quotative THEME from sections 2 and 3. I additionally show here that smuggling does not pose an issue for probing search algorithms (Ke 2019; Chow 2022; Branan & Erlewine 2022; Milway 2023). Finally, I connect smuggling to a specific understanding of grammatical voice, unifying the above examples in (107)-(110) as non-active inverse voice constructions (Givón 1994; Storment 2025).

4.1 Smuggling and the SMT

Here, I maintain a strong notion of consistency and uniformity in the projection of thematic arguments: configurational Theta-theory (Gruber 1997; Gruber & Collins 1997; Hale & Keyser 2002), the Theta- Criterion (Chomsky 1986), and, more recently, the Argument Criterion (Collins 2024). These are, in short, underlying principles that motivate a descriptive generalization such as the Uniformity of Theta-Assignment Hypothesis (UTAH) (Baker 1988). I adopt this general framework of argument structure in accordance with the SMT (Berwick & Chomsky 2016; Chomsky & Gallego & Ott 2019; Chomsky 2021; Chomsky et al. 2023; Chomsky 2024), which is defined below.

(111) *Strong Minimalist Thesis* (Berwick & Chomsky 2016:94)
The optimal situation would be that UG reduces to the simplest computational principles which operate in accord with conditions of computational efficiency. This conjecture is sometimes called the Strong Minimalist Thesis (SMT).

As such, we can simply and consistently define the Theta-roles of arguments for our purposes as follows: THEME the first argument externally merged within the VP, AGENT is the argument externally merged in Spec,vP (which is located above VP), and a GOAL is the argument externally merged in between a THEME and an AGENT.

The alternative conceptualization of argument structure, which I do not adopt here, relies not on any configurational structural means of Theta-role assignment, but on a powerful notion of formal semantics capable of connecting arguments in any position to any thematic role of a predicate (Heim & Kratzer 1998; Pylkkänen 2008; Bruening 2013; Myler & Mali 2021, a.o.). I do not adhere to such semantic notions of argument structure here as they are incompatible with a SMT-compliant formulation of how arguments are introduced. This is summarized with the Duality of Semantics principle of the SMT (Chomsky 2021), defined here.

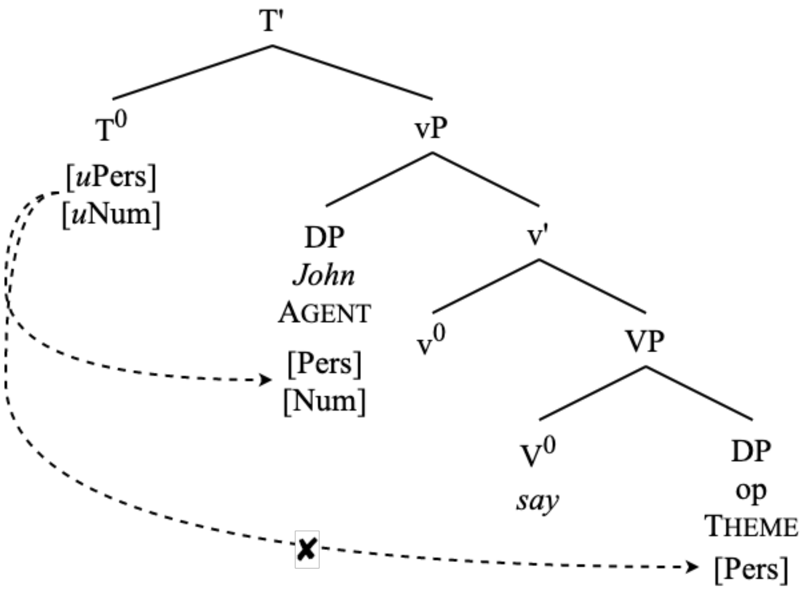
(112) *Duality of Semantics* (Chomsky 2021:18)
EM [External Merge] is associated with θ-roles and IM [Internal Merge] with discourse/information-related functions.

In the SMT, there is only Merge (ideally *at all*, minimally as a structure-building operation). As such, heavily incorporating tools of formal semantics into the explanation of argument structure building and theta-role assignment – and, indeed, attributing any explanatory power in the grammar to the lambda calculus – is contrary to the goals of SMT as it does not conform to the "simplest computational principles which operate in accord with conditions of computational efficiency" (111).

Another issue with the aforementioned semantic approaches to argument structure is that they raise "immediate and serious questions of restrictiveness" (Collins 2024:156) in the domain of argument structure but also beyond. For more discussion of this issue of restrictiveness, as well as more issues with these accounts overall, I refer the reader to chapter 10 of Collins (2024).

The framework of argument structure used in this paper is a Merge-based approach (Collins 2024), and this approach informs the need for smuggling and the general analysis of QI clauses. I have thus far presented a formulation of argument structure in which AGENTs are uniformly introduced in a position (Spec,vP) above that of THEMEs (VP). Returning to the examples in (107)-(110), but of course focusing mainly on QI, these sentences now have a problem of minimality (a problem which may have initially served as motivation to abandon an SMT-compliant Merge-based approach to argument structure). How does the THEME end up in an A-position (see section 3) over the AGENT, assuming the featural specification of AGENT is such that it blocks the probing of the THEME?

(113) Minimality problem of QI clauses: '*"Hey," said John*.'



This problem, however; is surmountable. Our solution comes in the form of smuggling (Collins 2005; Belletti & Collins 2021; Storment 2025, a.o.). Smuggling refers to the movement of a given element within a larger constituent, thus making that element a viable goal for processes that would normally be blocked by an intervener, as the larger consistent moves to a position above the intervener. See the following definition.

(114) *Smuggling* (Collins 2005:97)

Suppose a constituent YP contains XP. Furthermore, suppose that XP is inaccessible to Z because of the presence of W (a barrier, phase boundary, or an intervener for the Minimal Link Condition and/or Relativized Minimality), which blocks a syntactic relation between Z and XP (e.g., movement, Case checking, agreement, binding). If YP moves to a position c-commanding W, we say that YP smuggles XP past W.

Formally, this looks as follows.

(115) a. $[\mathrm{Z}_{[u\mathrm{F}]} \ldots [_{\mathrm{BP}}\ \mathrm{W}_{[\mathrm{F}]} \ldots [_{\mathrm{YP}}\ \mathrm{XP}_{[\mathrm{F}]}]]]$

b. $[\mathrm{Z}_{[u\mathrm{F}]} \ldots [[_{\mathrm{YP}}\ \mathrm{XP}_{[\mathrm{F}]}]_i \ldots [_{\mathrm{BP}}\ \mathrm{W}_{[\mathrm{F}]} \ldots [t_i]]]]$

The solution to the minimality problem of these inverted clauses in a Merge-based system, then, is Merge. Smuggling is simply two applications of internal Merge in which the first application facilitates the second. As such, there is nothing about smuggling that should be ruled out in a strict sense, though its distribution does bring up some interesting implications that I discuss in the remainder of this section.

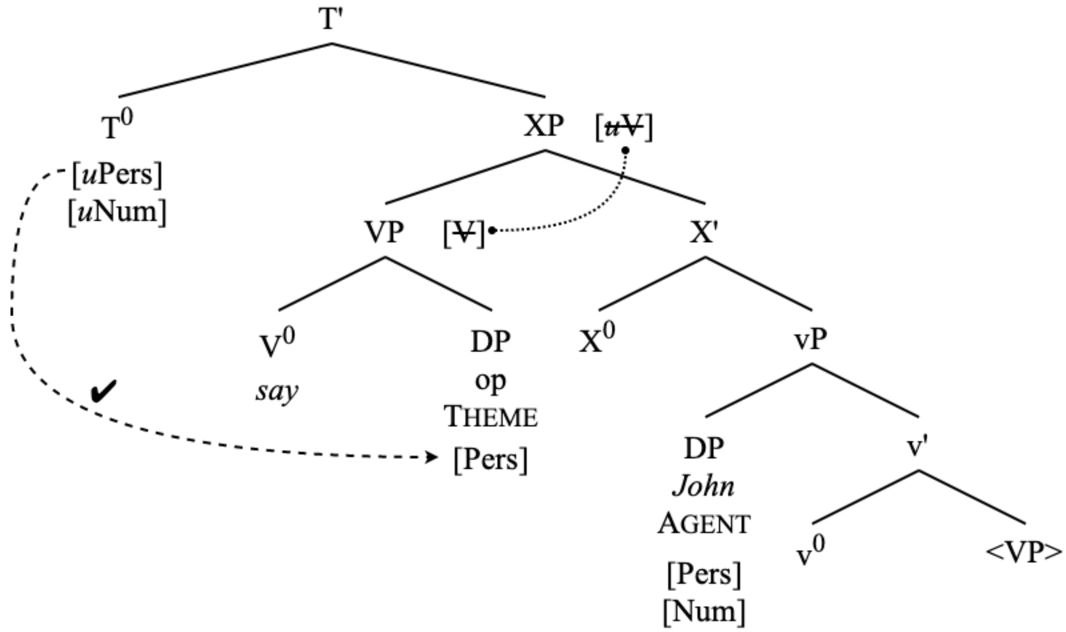
For QI specifically, smuggling is implemented as follows (Collins 2003; 2024; Storment 2025): In its base position, the direct speech report THEME is not a viable candidate for A-movement to the position of structural subject, assuming that such a movement is driven by phi-probing by a functional head such as T^0 (see section 3.1) due to the intervention of the AGENT. See (113).

Following (114), moving a maximal projection containing the THEME – but not the AGENT – to a position above the AGENT circumvents the intervention and frees up the THEME to participate in new A-dependencies. Section 2 has empirically identified the maximal projection that moves as VP, and this movement is driven by a functional projection whose featural specification is such that it attracts the VP for movement to its specifier, thus facilitating A-movement of the quotative THEME (section 3).

For a further discussion on the featural composition of this head, it is sufficient to say that this smuggling projection's head, which is merged directly above vP, contains an unvalued feature that verbal functional projections share—say, [*u*V]—that attracts the closest eligible verbal projection for movement to its specifier. Since this head's vP complement could not undergo complement-to-specifier movement due to anti-locality (Grohmann 2003), the complement of the vP—which is VP—is selected for movement to the specifier of the smuggling projection, since it also bears a [V] feature[7]. These considerations are borne out in (116) below. Recall that, according to section 3.5, the quote itself is actually not present in the syntactic derivation of the utterance, only the operator. This is evidenced by the fact that the quote may be totally absent from QI clauses (see again the *Says who?!*-style examples in (42)).

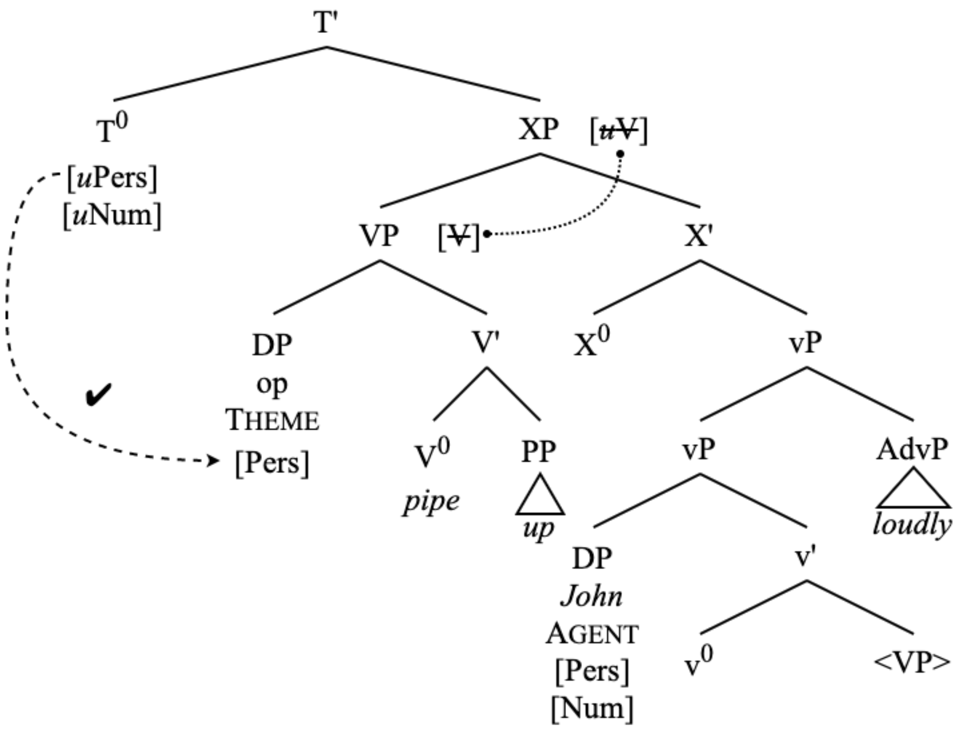
[7] For further comments on this featural motivation, I direct the reader to chapter 3 of Storment (2025). For explanations motivating this movement via the labeling algorithm, I direct the reader to Stegovec (2024) and to Blümel & Collins (2025).

(116) Smuggling circumventing the issue of minimality in QI: '*"Hey," said John*.'



In this structure, it is clear that only the VP is moving to a position above the external argument AGENT, thus deriving the empirical facts from section 2. Elements such as control clauses and English particles as VP-internal elements thus always come to precede the AGENT in these QI clauses, and elements that are external to the VP, such as vP-level adjuncts, do not precede the AGENT. Furthermore, the Theme is now in a position where it is accessible to the phi-probe on T^0.

(117) Structure containing both a VP-internal particle and vP-level adjunct.
'*"Hey!" piped up John loudly*.'



This derivation brings up a couple points that deserve explanation, though: namely, how we can be sure that a probing search algorithm will reliably find the THEME over the AGENT in (116) for A-movement, and what is the nature of the functional head that facilitates the VP movement.

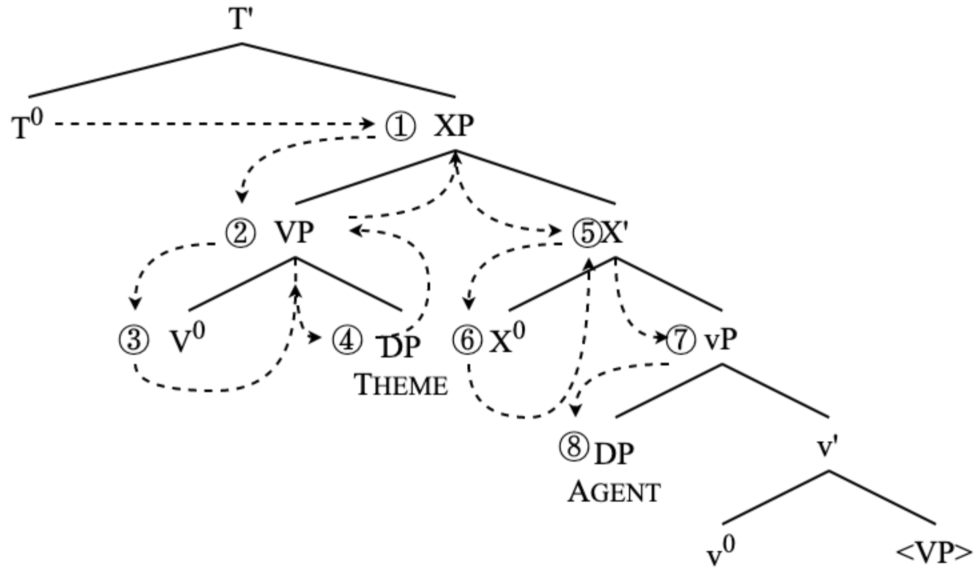
4.2 Probing algorithms

It is not immediately clear how it is ensured that a probe would always find the THEME over the AGENT in (116), given that the two elements no longer stand in an asymmetrical c-command relation, which is how closeness is typically defined in Minimalist grammars (Chomsky 1995; Ura 2000), although it is the case that a nonphasal (that is: transparent for probing) constituent containing the THEME does asymmetrically c-command the AGENT.

It is important to note that, even without direct asymmetrical c-command, a probe's search algorithm will find the THEME before the AGENT. No matter which parsing algorithm is posited to underly an operation such as minimal Search (Chomsky 2000; 2001), the probe will find the THEME over the AGENT here. I focus now on Breadth-First Search (BFS) and Depth-First Search (DFS) parsing algorithms.

In a DFS algorithm, which "starts at the root of an object and searches to a terminal node before backtracking" (Milway 2023:13), the probe will search the entirety of the VP containing the THEME before moving on. This is schematized below.

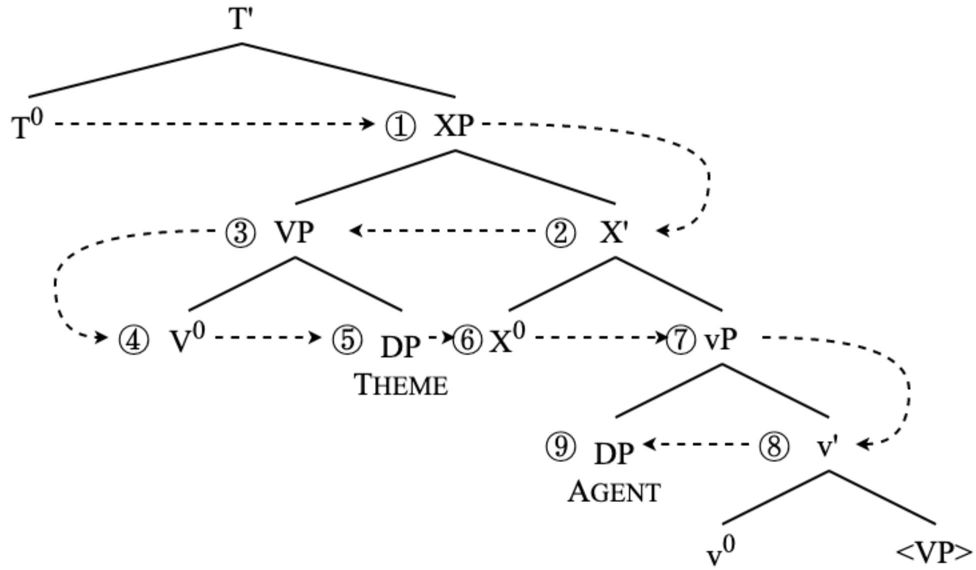
(118) DFS search algorithm in smuggling derivation.



To ensure here that the probe would search the VP before moving on to the domain of X' in the structure above, we will need to say a bit more about the DFS search path. Assuming that the parser cannot "see" information such as left-right linearization, it must be the case that the search algorithm, when faced with the choice of parsing two phrasal sisters, searches the domain of (non-phasal) maximal projections before it searches the domain of non-maximal projections.

It is certainly not unheard of to stipulate DFS for probing in Agree operations (Preminger 2019; Ke 2019; Chow 2022; Branan & Erlewine 2022, a.o.). Branan & Erlewine (2022:6-8) specifically mention DFS as necessary for smuggling derivations, though it should be pointed out that, even a BFS algorithm, which "searches neighbour nodes before proceeding lower in the tree" (Milway 2023:14), a search would still find the THEME before the AGENT in (116). See the following derivation.

(119) BFS search algorithm in smuggling derivation.



Following the aforementioned sources, though, I maintain that something like the DFS search path shown in (118) is advantageous for smuggling derivations as extra structure such as a control clause within the VP would change the results of the BFS search path. Another issue with the BFS search path as shown below is that it is necessarily ordered left-right, while trees are inherently unordered objects.

These considerations on the probing cause us to arrive at the following generalization.

(120) *Closeness Without C-Command*
For a configuration in which a probe on a head H^0 asymmetrically C-commands two nonterminal nodes α and β, and α is non-phasal, and α asymmetrically C-commands β, the probe will search the entirety of the structure dominated by α for a relevant goal before it searches the structure dominated by β.

One potential objection to smuggling (which I see as related to some question such as "Why is a search path of the kind shown in (118) not always possible?") is that extraction from a specifier often is not possible at all. To this, I say that we do not ultimately want a descriptive generalization—such as that extraction from specifiers is highly restricted—stated as an independently motivated condition in the grammar without sufficient evidence to do so, in line with the SMT. A fundamental distinction of this kind between complements and specifiers – both of which are added to a structure via Merge – is not ideal under any version of Minimalist theorizing, a goal of which is to "eliminate unnecessary distinctions" (Roberts 2010:5).

The question that first deserves careful consideration before smuggling could be ruled out on these grounds is: why is it the case that most specifiers cannot be extracted from, and should we expect that to hold in cases of smuggling? Generalizations such as the Left Branch Condition (Ross 1967) or the Specifier Constraint (Chomsky 1973; Corver 2017) ultimately apply to the ban on extraction from specifiers which are phasal, and thus opaque to probing (e.g., DP, PP).

Given that VP is not phasal, extraction from a VP that has moved into a specifier position should not be expected to be an issue. If recent work suggesting that phasality is ultimately reducible to relativized minimality for some feature (e.g., [ϕ]) is on the right track (Halpert &

Zeijlstra 2024; Keine & Zeijlstra 2024), then constraints on extractions from specifiers are explained by maximal projections in a Spec position bearing a valued instance of [ϕ] which renders them opaque for further probing. As VP is not valued for phi-features, extraction from a VP in a Spec position is not a violation of anything.

Ultimately, a descriptive "ban" is not sufficient grounds to rule out smuggling. In fact, there seems to be no aspect of these sequential applications of internal Merge that are ruled out in a Minimalist framework under the SMT. For related discussions regarding freezing, smuggling, and labeling, see Bošković (2020).

4.3 Smuggling and voice

Following Collins (2005; 2024) and Storment (2025), I adopt a conceptualization of grammatical voice of which smuggling is at the core. See the following definition.

(121) *Voice* (Storment 2025:19-20)
An information-structurally marked configuration in which a given thematic argument is mapped onto an A-position that would not be accessible to the same argument in a standard unmarked configuration. This alternation is dictated by the presence of a functional projection VoiceP that mediates between the positions in which arguments of a predicate are externally merged (i.e., theta-positions) and A-positions that arguments move to via internal Merge.

As such, I identify the smuggling projection—XP in (116)—as VoiceP. This conception of VoiceP is a notable departure from the view of $Voice^0$ as an applicative head which introduces the external argument, a view which currently has considerable traction in the field (Kratzer 1996; Pylkkänen 2008; D'Alessandro et al. 2017; Oxford 2023, a.m.o.) and is functionally equivalent to the notion of vP used in this paper.

There are several theoretical and empirical advantages to separating the head that encodes grammatical voice alternations from the head that introduces the external argument, which have been clearly laid out elsewhere. For an exhaustive discussion of these advantages, I direct the reader to chapter 8 of Collins (2024) and chapter 1 of Storment (2025). The ideas expressed in those works are echoed elsewhere (e.g., Merchant 2013:78-79; Myler & Mali 2021:3).

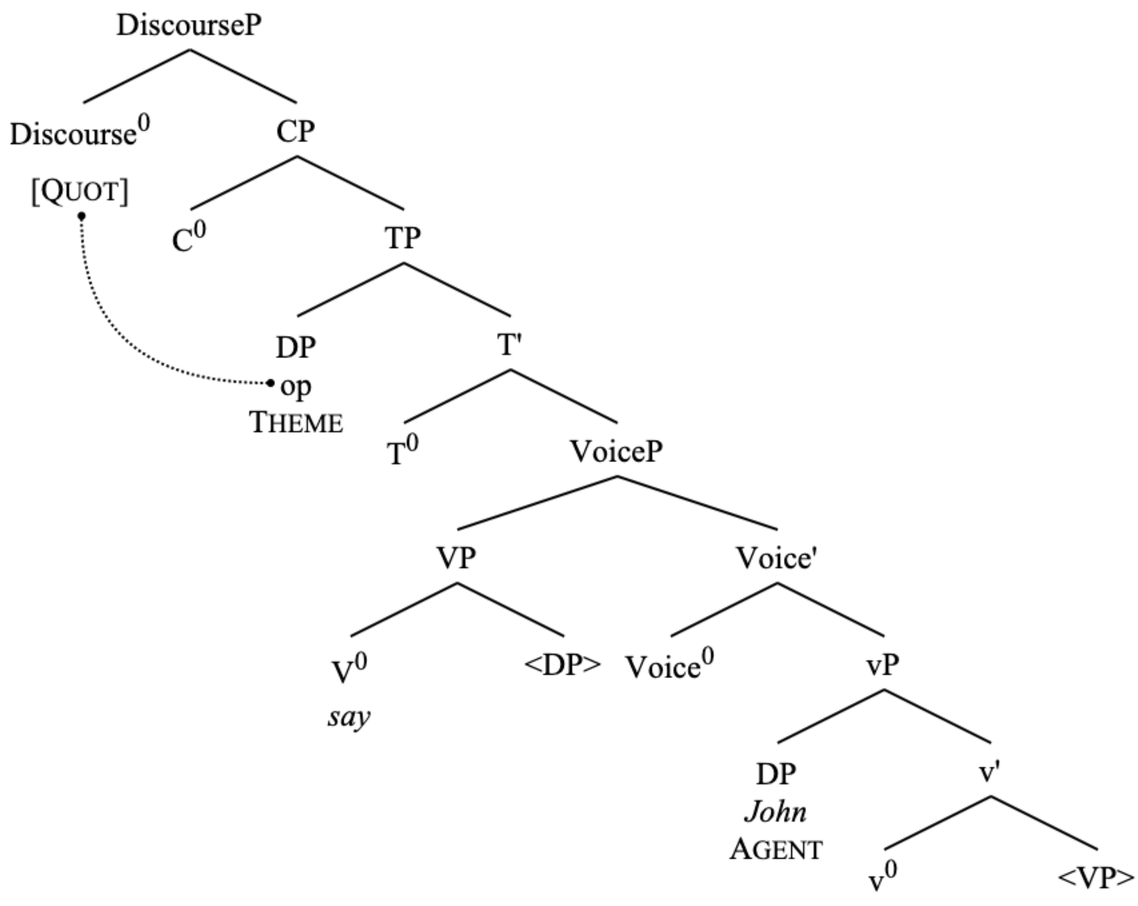
For the purposes of this paper, it is sufficient to say that grammatical voice alternations and the introduction of an external argument to a predicate are clearly conceptually distinct empirical phenomena, and one will find numerous issues in developing a cohesive theory of grammatical voice if the two are conflated. The voice and the valence of a predicate are separate alternations. Analytically, separating voice from valence frees us up to create a cohesive theory of grammatical voice in which voice alternations that do not affect the valence of a predicate can be uniformly analyzed. Seemingly disparate constructions that can all be conceived of as voice phenomena have been analyzed as involving smuggling, including passives (Collins 2005; 2024), dative shift (Collins 2021; 2024), causatives (Belletti 2017; Belletti & Collins 2021), middles (Gotah 2024), and inverse voice constructions (Shlonsky 2024; Storment 2025).

It is possible, though, to analyze voice alternations which do not affect the valence of a predicate under the external-argument-introducing view of VoiceP. Oxford (2023) does exactly this, keeping VoiceP central to the voice alternations and not making use of an additional stipulative functional projection such as PassP (see Bruening 2013). The issue with this kind of analysis, though, is that it necessitates theoretical stipulations such as equidistance in a search algorithm (Oxford 2023:331) and multiple specifiers (Oxford 2023:338). If we can do without such theoretical complications, as the analysis presented here does, then I consider that a plus, as it is more closely in accordance with the principles of SMT.

Further regarding the smuggling conception of VoiceP's compatibility with the SMT, let us again recall the Duality of Semantics (112). External Merge is associated with the thematic status of arguments, and internal Merge (i.e., movement) is associated with the discourse and information-structural status of those arguments. The smuggling VoiceP is not an argument-introducing head; it does not involve external Merge of any argument. Instead, it only facilitates smuggling via internal Merge. Given that inverse voice alternations do not affect the thematic status of the arguments involved in the predicate (an SVO quotative clause involves an Agent and a Theme, as does a QI clause), and that voice alternations encode specific information-structural configurations related to the discourse-prominence of certain arguments (Givón 1994), it naturally follows from the SMT internal—not external—Merge should be at the core of grammatical voice alternations.

Given this discussion (if it was not already clear), QI constructions in this framework can be thought of as inverse voice constructions in the languages I discuss in this paper. As such, the complete derivation of QI as smuggling is shown here.

(122) Full derivation of QI clauses: '*"Hey," said John*.'



Empirical support is provided for this claim further on by showing the similarity of QI to other types of inverted clauses (namely locative inversion) in English, Setswana, and other related Bantu languages.

5 The Transitivity Constraint

QI clauses in both English and Setswana are subject to the transitivity constraint, which blocks an indirect object GOAL DP from appearing in QI clauses (Collins & Branigan 1997), even with predicates that normally allow an indirect object DP.

(123) John warned Mary, 'There is danger ahead.'

(124) Thabo o tlhagisitse Neo, 'Go na le mathaithai kwa pele.'
Thabo SM.3SG warn Neo 'SM17 be with puzzles 17LOC ahead'
'Thabo warns Neo, "There is danger ahead."'

(125) *'There is danger ahead,' warned Mary John.

(126) *'Go na le mathaithai kwa pele,' go tlhagisitse Neo Thabo.
'SM17 be with puzzles 17LOC ahead' SM17 warn Neo Thabo
int: '"There is danger ahead," warns to Neo Thabo.'

Reversing the positions of the AGENT and the GOAL DPs do not improve the sentences.

(127) *'There is danger ahead,' warned John Mary.

(128) *'Go na le mathaithai kwa pele,' go tlhagisitse Thabo Neo.
'SM17 be with puzzles 17LOC ahead' SM17 warn Thabo Neo
int: '"There is danger ahead," warns Thabo to Neo.'

The only way around the transitivity constraint in QI clauses is to express the GOAL through means other than a DP argument. In English, this takes the form of a dative PP GOAL, and in Setswana, this is done with an adjunct clause.

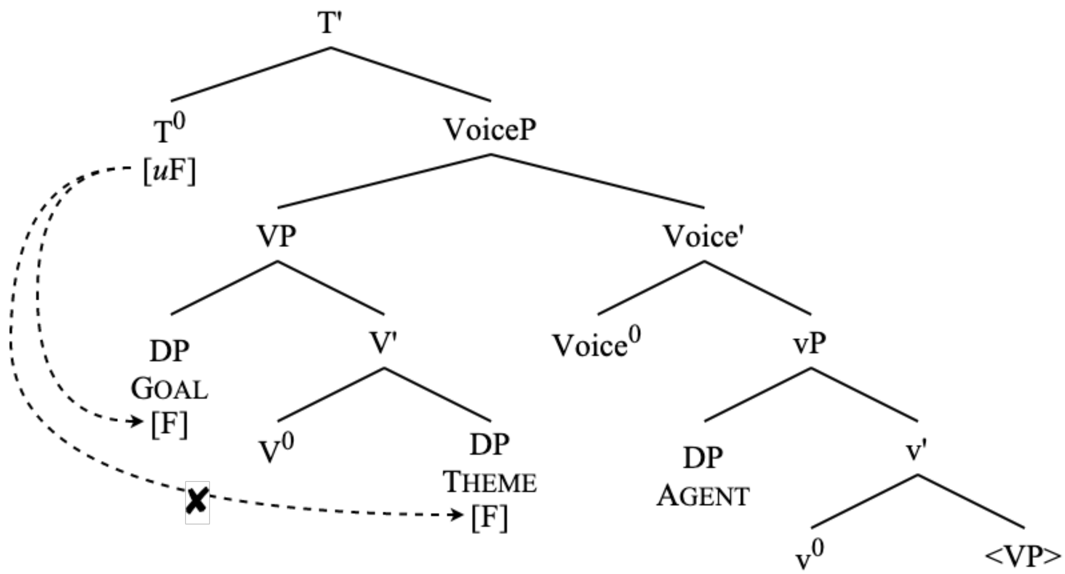
(129) 'There is danger ahead,' warned John to Mary.

(130) 'Go na le mathaithai kwa pele,' go tlhagisitse Thabo a tlhagisa
SM17 be with puzzles 17LOC ahead' SM17 warn Thabo SM.3SG.PST warn
Neo.
Neo
'"There is danger ahead," warned Thabo, warning Neo.

It is not immediately clear why the technology of inversion via smuggling should not be available depending on how many DP arguments the verb takes. I argue here that a constituent containing both a GOAL and a THEME DP does not lead to a viable result when smuggling occurs because both smuggled arguments are unable to be licensed in such a configuration.

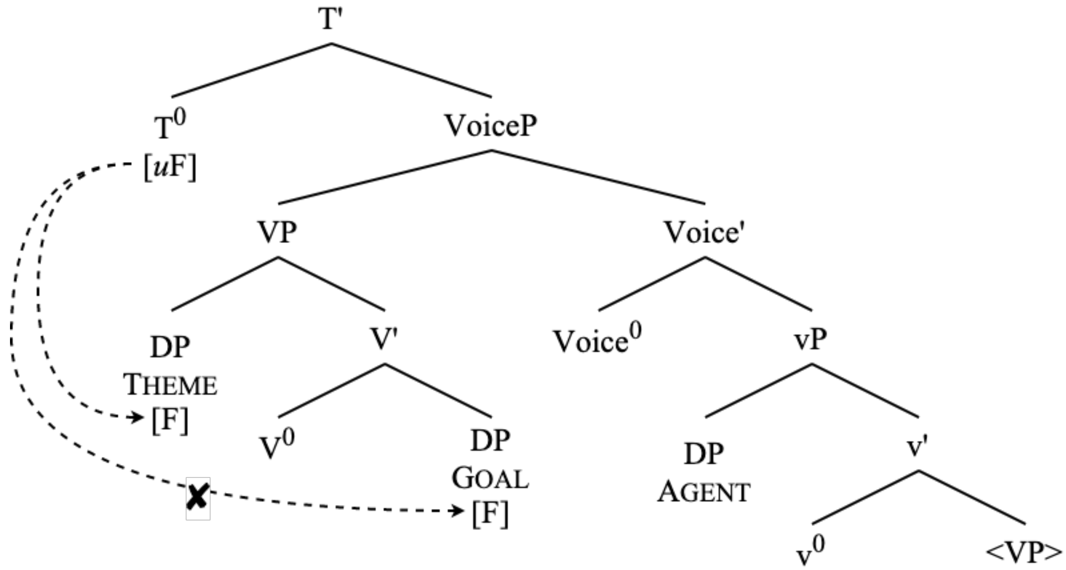
Assuming that an argument smuggled via Spec,VoiceP must be licensed in the Infl domain (i.e., it does not end up getting licensed in its base position) via phi-agreement with a functional head (here T^0), a derivation in which a VP that moves to Spec,VoiceP contains more than one argument DP (which I represent simplistically as two arguments merged within VP) poses an issue for the licensing of those two internal arguments. We can contrast such cases with how subject arguments are normally licensed in canonical non-inverted configurations in English, in which an agentive structural subject is licensed by T^0 (thus receiving nominative Case) and a thematic object is licensed by v^0 (thus being marked accusative) (Chomsky 2000; 2001). In other words, 'normal' subject argument licensing involves one argument being licensed by T^0.

(131) T^0 fails to license a smuggled THEME in the presence of a GOAL intervener.



Even in a situation in which the THEME DP could somehow become the closest suitable goal to the phi-probe on T^0, thereby solving an issue of minimality, the DP GOAL argument would then be unable to licensed as well, thus causing a Case filter violation (i.e., the lower argument is unable to be valued for Case by being probed by a Case-licensing functional projection).

(132) T^0 fails to license a smuggled GOAL in the presence of a THEME intervener.

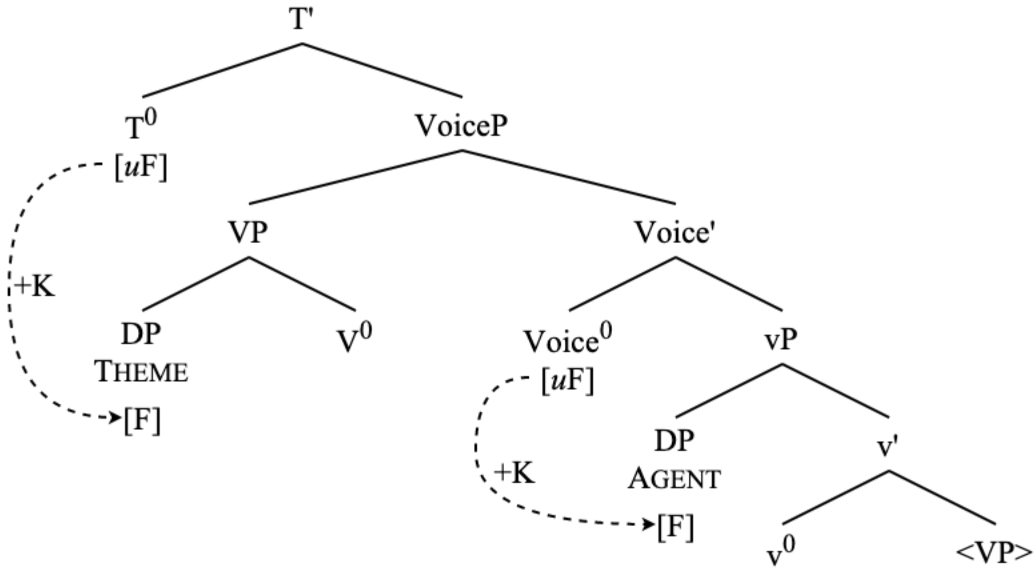


Connecting licensing by T^0 (and subsequent movement of the licensed argument to Spec,TP) here to nominative Case assignment (Chomsky 2000; 2001), it is clear that English is not such a language in which nominative Case can be assigned to multiple arguments with distinct thematic roles. The derivation in (131) contains an unlicensed THEME while (132) contains an unlicensed GOAL. Even if phi-agreement could obtain with the GOAL argument in (132) via defective circumvention, the argument would still be unable to be licensed as that would require movement to Spec,TP which is not available as the THEME would have already moved there.

Given that, according to the diagnostics from section 2, the non-DP GOALs in (129) and (130) are not located within the constituent that is smuggled to Spec,VoiceP, the expression of a GOAL in an adjunct clause or inside of a VP-external PP does not cause the same issues for licensing. The GOAL arguments in those cases are licensed internal to the PP and internal to the adjunct clause, respectively. I would like to remain somewhat agnostic here as to whether these non-DP GOAL constituents are base-generated outside of the VP or whether or not they evacuate the VP (Collins 2003) before smuggling occurs (though the uniform projection of the thematic arguments adopted in section 4 certainly suggests that it would be advantageous for a GOAL of any kind to be base-generated VP-internally), because, ultimately the effect of the goal not serving as a blocker or blockee within the smuggled VP is the same either way.

Following this discussion of the licensing of the internal argument and the inability of multiple internal argument DPs to be licensed in QI sentences, one may wonder how the in-situ external argument AGENT is licensed in QI, given that it certainly cannot be the argument that is licensed in Spec,TP. Here, following Collins (2005; 2024), I assert that Voice^0 must be able to license the external argument in addition to facilitating movement of the VP. As such, we end up with the following derivation regarding the licensing of the arguments in QI. The fact that the AGENT is not being licensed by T^0 in this situation can explain how the postverbal AGENT in QI clauses is not clearly associated with nominative Case. See how Case licensing of arguments normally works in a licit QI derivation, with [+K] indicating the Case valuation of an argument by undergoing Agree with a relevant functional head (again, see Chomsky 2000; 2001).

(133) THEME licensed by T^0, AGENT licensed by $Voice^0$



The analysis of the transitivity constraint presented here relies on the necessary licensing of thematic arguments. In other words, these arguments are subject to the Case filter (Chomsky 1981; 1986). This analysis, then, and the ways in which the behavior of Setswana QI mirrors that of English, lends support to the view that nominal arguments must indeed be Case-licensed in Bantu (Carstens 2005; Halpert 2012; 2017; Pietraszko 2020).

This explanation of the transitivity constraint additionally predicts that any inverse voice construction with more than one internal argument should be impossible, which is indeed the case for locative inversion in English, as well as locative inversion and existential clauses in Setswana (Demuth & Mmusi 1997; Diercks 2017). In other words, any inversion construction that involves smuggling shows the transitivity constraint. See the following.

(134) a. *Into the room walked Mary her dog.
b. *Into the room walked her dog Mary.

(135) *Go et -ela basimane koko. (Demuth & Mmusi 1997:14)
SM17 visit-APPL boys grandmother
int: 'There are boys visiting the grandmother.'

Although, locative inversion with transitive verbs has been reported to be possible in some Bantu languages such as Otjiherero (Marten 2006) and Digo (Diercks 2012; Nicolle 2013) (Diercks 2017), which seems to stand as evidence that the transitivity constraint is not a universal in all inversion constructions in all languages[8]. The question of the universality of the transitivity

[8] Interestingly, there seems to be a generalization in Otjiherero and Digo that a THEME object must come after the AGENT postverbal DP in transitive locative inversion clauses.

(iv) pò- ngàndá pé- térék-à òmú-kázéndú ònyàmà. (Otjiherero, Marten 2006:117)
16LOC-house SM16-cook-FV 1- woman meat
'At home the woman cooks meat.'

constraint and how it is resolved presents a very interesting line of investigation for future study. For the purposes of QI in English and Setswana, I have shown that the transitivity constraint naturally follows from conditions on the Case-licensing of arguments and the structure of the smuggling derivation.

6 Parallels with locative inversion

Throughout the paper, I have mentioned the parallels of QI to locative inversion (136) in both English (a) and Setswana (b).

(136) a. Down from the wall leapt Gimli.
b. Mo lefatshe-ng go fula dikgomo. (Demuth & Mmusi 1997:5)
18LOC country-LOC SM17 graze cattle
'In the country graze cattle.'

The parallels between locative inversion and QI in these languages are significant enough to warrant independent discussion, as the mechanism of inversion via smuggling via VoiceP that uniformly account for QI in two unrelated languages is also sufficient to account for locative inversion in those languages.

I do not replicate the tests here in the interest of brevity, but it is the case that the relevant constituency tests shown in section 3 all indicate VP-movement to a position above the external argument in locative inversion clauses as well. For a detailed discussion of these tests for locative inversion, see chapter 3 of Storment (2025).

Additionally, verbal agreement in locative inversion functions just as it does for QI in English (Storment 2025) and in Setswana (Demuth & Mmusi 1997; Creissels 2011; Diercks 2017). The facts surrounding the conjoint/disjoint alternation in Setswana are also the same for locative inversion; namely, that the disjoint form is not possible (Creissels 2011:fn. 9).

Locative inversion is additionally compatible with raising in English (see section 3.3.1) and in Setswana as well. See Demuth & Mmusi (1997), Creissels (2011) for classic raising, and see the following examples for the aspectual auxiliary raising shown in section 3.3.2.

(137) Seabelo o setse a tsene mo kamore-ng.
Seabelo SM.3SG already SM.3SG walk 18LOC room -LOC
'Seabelo already walked into the room.'

(v) Muho-ni pha- na- heka atu madzi. (Digo, Diercks 2012:267)
River-16LOC SM16-CONT-draw people water.
'At the river people are drawing water.'

These sources do not identify a THEME > AGENT word order as ungrammatical, but all of the examples given (as well as those in Diercks 2017) show a AGENT > THEME word order. This word order restriction, if it holds, would be an indication that the THEME in these transitive locative inversion clauses is not located inside of the constituent that raises for smuggling. It could be the case that the THEMEs in *(iv)* and *(v)* are actually in a position similar to the GOAL arguments in (129) and (130). As such, these data are not necessarily incompatible with a smuggling analysis of locative inversion and could very well ultimately conform to the transitivity constraint and strategies for circumventing the transitivity constraint presented in this section. Independent evidence is needed to confirm.

(138) Mo kamore-ng go setse go tsene Seabelo.
18LOC room -LOC SM17 already SM17 walk Seabelo
'Into the room already walked Seabelo.'

As such, the derivation of locative inversion clauses can be uniformly analyzed as involving smuggling, just the same as QI clauses in both languages. This expands the technology of inversion via smuggling to yet another constructions, and it expands the typology of inverse voice phenomena in both English and Setswana to include locative inversion.

This view of locative inversion is advantageous for several reasons. First of all, it needs not rely on the idea that locative inversion predicates are underlyingly unaccusative in all cases (Bresnan & Kanerva 1989; Bresnan 1994; Culicover & Levine 2001; Doggett 2004; Rizzi & Shlonksy 2006; Diercks 2017, a.o.), even when the verbs involved are not normally underlyingly unaccusative. To my knowledge, it has never been argued that QI verbs must be unaccusative, which shows that unaccusativity is not necessary in order to derive an inverted word order. In fact, the same technology of inversion via smuggling driven by VoiceP can apply to languages in which locative inversion is only possible with unaccusative verbs, such as Chichewa (Brensan & Kanerva 1989; Diercks 2017), as well as languages in which locative inversion is possible with both unergative and unaccusative verbs, such as English and Setswana.

Additionally, the smuggling account of locative inversion allows for a uniform analysis of locative inversion within Bantu with respect to the typology of "agreeing" versus "non-agreeing" locative inversion languages (Diercks 2017). See the following contrast between an "agreeing" language, Chichewa, and a "non-agreeing" language, Setswana.

(139) M -mitêngo mw- a- khal-a anyǎni. (Chichewa, Bresnan & Kanerva 1989:16)
18LOC-trees SM18-PRF-sit -IND baboons
'In the trees are sitting the baboons.'

(140) Mo kamore-ng {go/ *mo} tsene Seabelo.
18LOC room -LOC {SM17/*SM18} tsene Seabelo
'Into the room walks Seabelo.'

One existing analysis of the difference between agreeing and non-agreeing Bantu languages in locative inversion is that languages of the agreeing variety such as Chichewa actually involve the preverbal locative PP as the structural subject, whereas those of the non-agreeing variety like Setswana involve a null expletive that raises to the structural subject position, while the locative PP is externally merged in the left periphery as an adjunct (Creissels 2011). This view is contrasted with the idea that the locative phrase in agreeing Bantu languages such as Setswana occupies the structural subject position just as it does in agreeing Bantu languages, only that the Setswana locative subject does not trigger agreement in the same way that it would in a language like Chichewa (Demuth & Mmusi 1997).

In the view of non-agreeing Bantu locative inversion involving an expletive subject, locative inversion in Setswana is structurally similar to the analysis of QI presented in this paper, involving a null thematic argument that moves into the structural subject position and is bound by an element located above TP. Buell (2006; 2007) highlights the similarity between agreement in locative and quotative inversion in Zulu, a related language, and concludes that non-agreeing preverbal locative phrases – such as those in Setswana locative inversion – must be adjoined above TP (Buell 2007:117).

While the expletive view of Setswana locative inversion draws certain parallels with the analysis of QI, it does not seem entirely necessary. The reason that it is necessary to analyze the null quotative operator as occupying the THEME position in QI clauses is because the quote itself is not actually internal to the utterance in the way that a normal argument would be, a position that I have empirically defended in section 4.5. For locative inversion, there is no such reason to believe that the locative PP should be any more external to the argument structure than any normal THEME (e.g., locative subjects cannot appear to the right of an inverted clause, they are subject to the same well-formedness conditions as the rest of the utterance, PP-internal indexicals interact with the rest of the clause in a way that they do not for quotes, etc.).

Additionally, in Setswana (just as in English), a topic appearing in a clause-initial position in an embedded clause in which the AGENT precedes the verb is not possible.

(141) a. Ke akanya gore Thabo o tabog-etse kwa lebentlele-ng.
SM.1SG think that Thabo SM.3SG run -PRF 17LOC market -LOC
'I think that Thabo has run into the store.'
b. *Ke akanya gore kwa lebentlele-ng Thabo o tabog-etse.
SM.1SG think that 17LOC market -LOC Thabo SM.3SG run -PRF
int: 'I think that into the store Thabo has run.'

However, an embedded locative inversion clause is possible in Setswana[9].

(142) Ke akanya gore kwa lebentlele-ng go tabog-etse Thabo.
SM.1SG think that 17LOC market -LOC SM17 run -PRF Thabo
'I think that into the room has run Thabo.'

The asymmetry between (141b) and (142) suggests that the locative truly undergoes A-movement to its subject position in Setswana locative inversion, as, if the fronted locative in locative inversion were always a topic, then it should pattern like the topicalized locative in the Loc-S-V sentence in (141b).

An embedded QI clause is not possible with a preverbal quote, given that the quote itself is not located in a clause-internal position.

(143) *Ke akanya gore 'Dumela!' go bu -ile Seabelo.
SM.1SG think that 'hello' SM17 say-PRF Seabelo
int: 'I think that "Hello!" has said Seabelo.'

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[9] I have spoken with one native speaker of Setswana who does not like embedded locative inversion clauses as in (142). This specific point of variation mirrors what I have found in English as well. Some people accept the following, while others do not.

(vi) ?I think that in through the doors busted someone I have never met before.

In any case, though, (*vi*) is certainly an improvement over a sentence with an embedded fronted PP and no inversion.

(vii) *I think that in the through the doors someone I have never met before busted.

Interestingly, the quote appearing in a clause-final position in QI can improve the status of an embedded QI clause.

(144) ?Ke akanya gore go bu -ile Seabelo, 'Dumela!'
SM.1SG think that SM17 say-PRF Seabelo 'hello!'
'I think that said Seabelo, "Dumela!"'

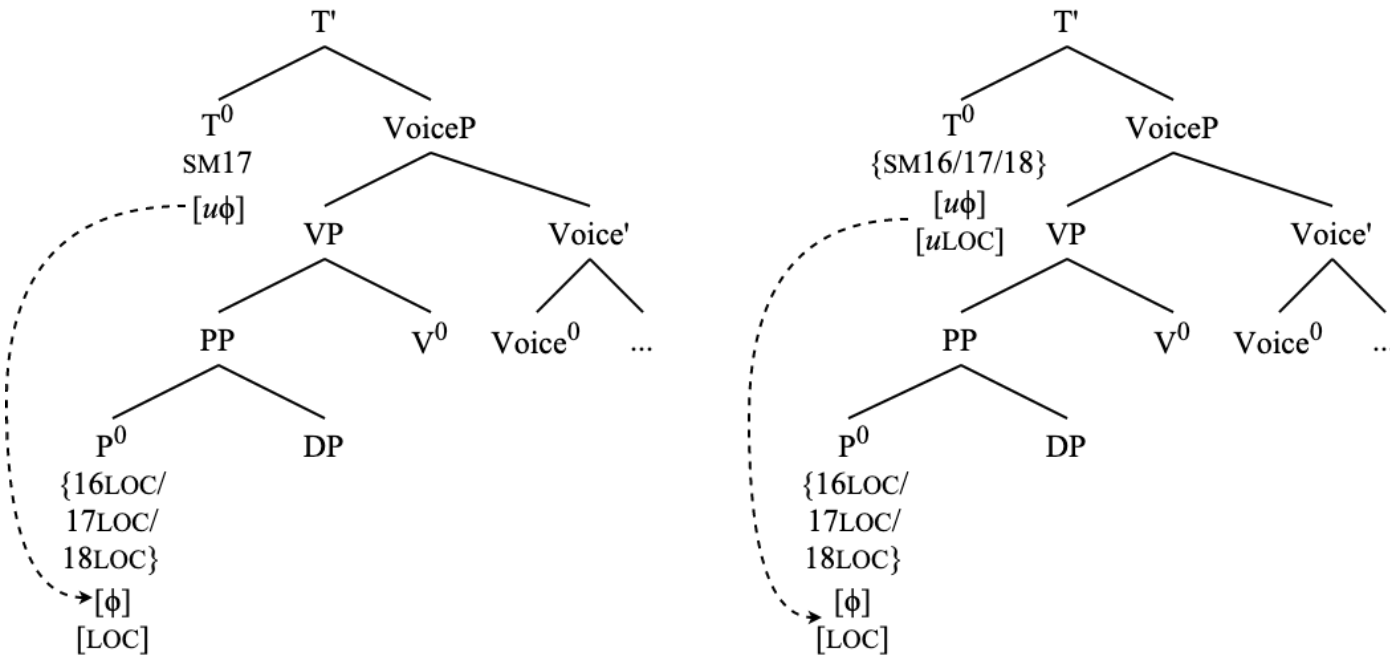
These facts regarding embedded QI clauses follow from the analysis of QI involving the smuggling of the null quotative operator to the position of the thematic subject. The data on the compatibility of locative inversion with embedded clauses then suggest that the locative constituent in Setswana locative inversion is not in the position of a topic, and is indeed the thematic subject. If it were in a topic position in locative inversion, we should not see an asymmetry between the LOC-V-S and LOC-S-V derivations, but we do. Thus, the analysis of a null expletive in Setswana locative inversion put forth by Buell (2006; 2007) and Creissels (2011) no longer seems particularly advantageous. Motivated by the data and discussion presented here, we now see a picture in which the locative subjects of locative inversion clauses in both agreeing and non-agreeing Bantu languages are derived in the same way that the quotative operator comes to be in the structural subject position. The only difference, of course, is that Setswana and languages like it do not show distinct forms of locative agreement, which does happen in other Bantu languages. This difference does not merit a completely different analysis of locative inversion for each of these language types; rather, the locative can be analyzed as the thematic subject in Setswana (Demuth & Mmusi 1997)—in addition to agreeing Bantu languages—given that the phi-agreement probes for subject-verb agreement in these are parametrically featurally distinct.

The idea that phi-probes are relativized across languages to search for different features is in no way new, and some form of this idea has been expressed in many, many formalizations of Agree (Béjar 2003; Nevins 2007; 2011; Georgi 2013; Preminger 2014; Coon & Keine 2021; Deal 2024; Storment 2025, a.m.o.). As such, we can apply this idea as it pertains to the relevant distinction between agreeing and non-agreeing locative inversion Bantu languages in order to arrive at a minimalist explanation of this difference that does not rely on positing completely different conditions on the derivation itself, just as I have shown for the differences in agreement in QI clauses across English and Setswana in section 3.1.

Locative nominal prefixes are distinct from other noun class morphemes in Bantu given that they convey locative and deictic information. Additionally, derived locative nominal prefix forms are used to form locative argument and adjunct clauses (Cole 1955:97). In fact, the locative markers seen in fronted locative subjects (140) are not even noun class morphemes at all, according to Cole (1955); they are what Cole refers to as the "locative adverbial formatives" (p. 97). As such, instead of being specified for noun class, we can conceive of these locative phrases as minimally consisting of [ϕ] (as they are available as goals for A-movement) and some locative feature [LOC], with no specific noun class or number features.

As established in section 3, the phi-probe on T^0 in Setswana bears only an feature necessitating a search for [ϕ]. Let us assume that this is true for the other non-agreeing locative inversion Bantu languages. Additionally, we can state that agreeing languages such as Chichewa are minimally different in that they have a feature on T^0 triggering interaction with the feature [LOC] somewhere in the structure of the probe. Thus, a locative goal is only visible to the Setswana probe as being a deficient phi-bearing element (no number feature, no class feature), whereas the same goals are visible in the agreeing locative inversion Bantu language as also having their relevant locative feature, which results in locative subject agreement on the verb.

(145) Phi-probing in non-agreeing (left) and agreeing (right) locative inversion Bantu languages.



The phi-probe in a language like Setswana is thus not sensitive to the locative features of the locative element[10], while the phi-probe in a language like Chichewa is. The parameterization of features on probes in a given language (the "relativization" (Georgi 2013) of these features, so to speak) characterizes the differences among Bantu languages with respect to locative inversion as superficial distinctions that boil down to the featural specification of phi-probes, which means

[10] I have the locative element in both Setswana and agreeing Bantu LI languages as a PP in the tree, though an anonymous reviewer has pointed me to a literature which suggests that the category and/or underlying internal structure of the locative phrases in agreeing and non-agreeing LI Bantu languages may actually be distinct from one another (Bresnan & Kanerva 1989; Bresnan & Mchombo 1995; Zeller & Ngoboka 2018). While such an analyses are certainly possible—and many very well be on the right track—I do not actually see such a distinction as crucial to the argumentation and analysis here. Even if the underlying structure of these locative phrases in each language type is distinct, that is no motivation to posit that they should be distinctly visible or invisible to an agreement probe. For instance, locative inversion in English (viii) is possible with both argument (a) and adjunct (b) locative phrases, as well as locative elements that are less-straightforwardly traditional PPs (c).

(viii) a. Onto the hot summer pavement stepped the cool cat.
b. With money comes problems.
c. Here lies Squidward's hopes and dreams.

Thus, any parametric differences in the underlying structure of the locative phrases across Bantu is not directly relevant for the analysis of agreement probing here, and, crucially, I do not believe that it should be—once again in the Minimalist spirit of eliminating unnecessary distinctions where possible (Roberts 2010:5). Ideally, these constructions should be underlyingly as similar as possible. Thus, I do not indicate such considerations in the structure in (145).

that completely separate derivations, technologies, or numerations are not needed to explain the same construction across languages which otherwise behaves quite similarly. This is the same mechanism by which I have explained the difference between agreement in English and Setswana across QI clauses.

As such, quotative inversion in English and Setswana, as well as locative inversion in English, Setswana, and other Bantu languages (regardless of agreement) are all uniformly analyzed as inverse voice constructions that involve smuggling of an internal argument to the structural subject position via VoiceP. Superficial differences in agreement in these constructions across languages are characterized by parametric differences in the featural composition of phi-probes, and thus pose no significant challenges to the uniform classification of these phenomena.

7 Conclusion

In this paper, I have shown in detail that the issues of minimality, A-movement, and agreement in quotative inversion constructions in two unrelated languages can uniformly be analyzed using principles of the SMT and a merge-based conceptualization of argument structure. The technology by which I have accomplished this relies primarily on inversion via smuggling, ultimately driven by a novel view of grammatical voice.

This analysis has sufficiently accounted for a variety of data concerning constituency, movement, and agreement in both Setswana and English. I have also accounted for language-specific data points such as the distribution of particles in English and the conjoint/disjoint alternation in Setswana, showing that these properties of QI also follow from the smuggling analysis. Additionally, I have shed light on the syntax of quotation in general, showing, as Collins & Branigan (1997) say, that quotes are best conceived of as external to the utterances along with which they typically appear.

Furthermore, I have demonstrated that the principles under which I have analyzed quotative inversion also lend themselves to an analysis of locative inversion as smuggling, an idea which is empirically supported in the same way. In so doing, I have shed light on the differences in agreement in locative inversion among certain Bantu languages. By extension, I have presented a uniform way of thinking about both locative inversion and quotative inversion in English and Bantu, accounting for the superficial differences in agreement across these languages. Ultimately, I believe that this technology can account for subject-object inversion in other Bantu languages (Marten & van der Wal 2015; Shlonsky 2024), though I do not show such cases here.

This work contributes an expanded typology of what counts as a voice alternation. I present here the idea that OVS word orders derived by A-movement in typically SVO languages can and should be thought of a inverse voice constructions. Given that smuggling can sufficiently account for other grammatical voice alternations such as passives (Collins 2005; 2024) and causatives (Belletti 2017), it is clear that these constructions that involve the promotion of an internal argument to a position of relative prominence above the external argument via A-movement should be cast in terms of grammatical voice, with smuggling driving the analysis.

There are many interesting aspects of QI sentences that I have not addressed here, such as the incompatibility with modals/auxiliaries, negation, and question formation in English, the interactions with Heavy-NP Shift in QI in English and Setswana, and the interactions between QI and passivization in both languages. I leave these issues open for future work, as I believe they are not consequential enough to the core technology of inversion via smuggling to address here. Additionally, I have left open the question of why both English and Setswana lack a more general inverse voice construction (i.e., subject-object inversion) despite having QI and locative inversion, which I believe is ultimately related to the deficiency of quotative/locative subjects. QI constructions remain relatively undescribed, and there is much work left to do on the behavior of QI crosslinguistically. The analysis I have put forth in this paper provides a framework in which to analyze the construction in different language varieites.

References

Angelopoulos, Nikos. 2025. Nominalization of clauses: The clausal prolepsis strategy. *Natural Language & Linguistic Theory* 43, 151-165. https://doi.org/10.1007/s11049-024-09618-4.

Bailyn, John. 2004. Generalized Inversion. *Natural Language & Linguistic Theory* 22(1), 1-50. https://doi.org/10.1023/B:NALA.0000005556.40898.a5

Baker, Mark. 1988. *Incorporation: A theory of grammatical function changing.* The University of Chicago Press.

Baker, Mark. 2003. Agreement, dislocation, and partial configurationality. In Harley, Heidi & Carnie, Andrew & Willie, MaryAnn (eds.) *Formal approaches to function in grammar*, 107-132. Amsterdam: Benjamins.

Baker, Mark. 2008. *The Syntax of Agreement and Concord*. Cambridge University Press.

Béjar, Susana. 2003. *Phi-syntax: A theory of agreement*. Doctoral dissertation. University of Toronto.

Béjar, Susana & Rezac, Milan. 2003. Person licensing and the derivation of PCC effects. In Pérez-Leroux, Ana Teresa & Roberge, Yves (eds.) *Romance linguistics: Theory and acquisition*, 49–62. Amsterdam: John Benjamins.

Bekeuma, Frits & Coopmans, Peter. 1989. A Government-Binding perspective on the imperative in English. *Journal of Linguistics* 25, 417–436.

Belletti, Adriana. 2017. Labeling (Romance) causatives. In Aboh, Enoch & Haeberli, Eric & Puskás, Genovea & Schönenberger, Manuela (eds.) *Elements of Comparative Syntax: Theory and Description*, 13-46. https://doi.org/10.1515/9781501504037-002.

Belletti, Adriana & Collins, Chris. 2021. *Smuggling in syntax.* Oxford University Press.

Berwick, Robert & Chomsky, Noam. 2016. *Why Only Us*. Cambridge, MA: MIT Press.

Blümel, Andreas & Collins, Chris. 2025. Smuggling and labeling theory. *Biolinguistics* 19, e17229. https://doi.org/10.5964/bioling.17229.

Bošković, Željko. 2021. On smuggling, the freezing ban, labels, and *tough*-constructions. In Belletti, Adriana & Collins, Chris (eds.) *Smuggling in syntax*, 53-95. Oxford University Press. https://doi.org/10.1s

Bresnan, Joan. 1994. Locative Inversion and Universal Grammar. *Language* 70, 72–131.

Bresnan, Joan & Kanerva, Jonni. 1989. Locative Inversion in Chichewa: A Case Study of Factorization of Grammar. *Linguistic Inquiry* 20(1), 1–50.

Bresnan, Joan & Mchombo, Sam. 1995. The lexical integrity principle: Evidence from Bantu. *Natural Language & Linguistic Theory* 13, 181-254.

Bruening, Benjamin. 2013. By phrases in passives and nominals. *Syntax* 16(1), 1-41. https://doi.org/10.1111/j.1467-9612.2012.00171.x.

Bruening, Benjamin. 2016. Alignment in Syntax: Quotative Inversion in English. *Syntax* 19(2), 111-155. https://doi.org/10.1111/synt.12121.

Buell, Leston. 2006. The Zulu Conjoint/Disjoint Verb Alternation: Focus or Constituency? *ZAS papers in linguistics* 43, 9-30.

Buell, Leston. 2007. Semantic and Formal Locatives: Implications for the Bantu Locative Inversion Typology. In Kula, Nancy & Marten, Lutz (eds.) *Bantu in Bloomsbury: Special issue on Bantu Linguistics*, 107-122. University of London Department of Linguistics.

Caha, Pavel. 2024. A default theory of default case. *Glossa* 9(1). https://doi.org/10.16995/glossa.16571

Carstens, Vicki. 2001. Multiple Agreement and Case Deletion: Against φ-incompleteness. *Syntax* 4(3), 147-163. https://doi.org/10.1111/1467-9612.00042.

Carstens, Vicki. 2005. Agree and EPP in Bantu. *Natural Language & Linguistic Theory* 23, 219-279. https://doi.org/10.1007/s11049-004-0996-6.

Chomsky, Noam. 1973. Conditions on Transformations. In Anderson, Stephen & Kiparsky, Paul (eds.) *A Festschrift for Morris Halle*, 232-286. New York: Academic Press.

Chomsky, Noam. 1981. *Lectures on government and binding: The Pisa lectures*. The Hague: Mouton.

Chomsky, Noam. 1986. *Knowledge of Language*. New York: Praeger.

Chomsky, Noam. 1995. *The minimalist program.* Cambridge, MA: MIT Press.

Chomsky, Noam. 2000. Minimalist Inquiries: The Framework. In Martin, Roger & Michaels, David & Uriagareka, Juan (eds.) *Step by Step: Essays on minimalist syntax in honor of Howard Lasnik*, 89-153. Cambridge, MA: MIT Press.

Chomsky, Noam. 2001. Derivation by phase*. Ken Hale: A Life in Linguistic*s. 1–52. https://doi.org/10.7551/mitpress/4056.003.0004, Chomsky, Noam. 2021. Minimalism: Where we are now, and where can we hope to go. *Gengo Kenkyu* 160, 1-41. http://doi.org/10.11435/gengo.160.0_1.

Chomsky, Noam. 2024. The Miracle Creed and SMT. In Greco, Matteo & Mocci, Davide (eds.) *A Cartesian Dream: A geometrical account of syntax in honor of Andrea Moro*, 17-40. Lingbuzz Press.

Chomsky, Noam & Seely, T. Daniel & Berwick, Robert & Fong, Sandiway & Huybregts, M.A.C. & Kitahara, & Hitatsugu & McInnerney, Andrew & Sugimoto, Yushi. 2023. *Merge and the Strong Minimalist Thesis*. Cambridge University Press.

Chomsky, Noam & Gallego, Ángel & Ott, Dennis. 2024. Generative Grammar and the Faculty of Language: Insights, Questions, and Challenges. Catalan Journal of Linguistics Special Issue, 229-261. https://doi.org/10.5565/rev/catjl.288.

Chow, Jeng Ki. 2022. A novel algorithm for Minimal Search. *Snippets* 42, 3-5.

Cole, Desmond. 1955. *An Introduction to Tswana Grammar*. Cape Town: Longman Penguin.

Collins, Chris. 1997 Field notes on Quotative Inversion in Setswana.

Collins, Chris. 2003. The Distribution of Particles in Quotative Inversion. Manuscript. Cornell University.

Collins, Chris. 2004. The Agreement Parameter. In Breitbarth, Anne & van Riemsdijk, Henk (eds.) *Triggers*, 115-136. https://doi.org/10.1515/9783110197433.115.

Collins, Chris. 2005. A smuggling approach to the passive in English. *Syntax* 8(2) 81-120. https://doi.org/10.1111/j.1467-9612.2005.00076.x.

Collins, Chris. 2021. A smuggling approach to the dative alternation. In Belletti, Adriana & Collins, Chris (eds.) *Smuggling in syntax*, 96-107. Oxford University Press.

Collins, Chris. 2023. Internet Searches as a Tool in Syntactic Research (second version). Blog post. https://ordinaryworkinggrammarian.blogspot.com/2023/01/internet-searches-as-tool-in-syntactic.html

Collins, Chris. 2024. *Principles of Argument Structure: A Merge-Based Approach*. MIT Press.

Collins, Chris & Branigan, Phil. 1997. Quotative inversion. *Natural Language & Linguistic Theory* 15(1), 1-41. https://doi.org/10.1023/A:1005722729974.

Coon, Jessica & Keine, Stefan. 2021. Feature gluttony. *Linguistic Inquiry* 52(4), 655-710. https://doi.org/10.1162/ling_a_00386.

Coopmans, Peter 1992. English Focus Constructions and the Theory of Grammar. *Language* 68(1), 206–210.

Corver, Norbert. 2017. Subextraction. In Everaert, Martin & van Riemsdijk, Henk (eds.) *The Wiley-Blackwell companion to syntax*. https://doi.org/10.1002/9781118358733.wbsyncom054.

Creissels, Denis. 1996. Conjunctive and disjunctive verb forms in Setswana. *South African journal of African linguistics* 16(4), 109-115.

Creissels, Denis. 2011. Tswana locatives and their status in the inversion construction. *Africana Linguistica* 17, 33-52.

Creissels, Denis. 2017. The conjoint/disjoint distinction in the tonal morphology of Tswana. In Van der Wal, Jenneke & Hyman, Larry (eds.) *The Conjoint/Disjoint Alternation in Bantu*, 200-238. De Gruyter Mouton. https://doi.org/10.1515/9783110490831-008

Creissels, Denis. 2023. The circumstantial forms of Tswana verbs and the notion of participle. *Studies in African linguistics* 52(13), 327-338.

Culicover, Peter, Polarity, inversion, and focus in English. In Westphal, German et al. (eds.) *Proceedings of ESCOL 1991*, 46-68.

Culicover, Peter & Levine, Robert. 2001. Stylistic Inversion in English: A Reconsideration. *Natural Language & Linguistic Theory* 19(2), 283-310. https://doi.org/10.1023/A:1010646417840.

D'Alessandro, Roberta & Franco, Irene & Gallego, Ángel. 2017. *The verbal domain*. Oxford University Press.

Deal, Amy Rose. 2024. Interaction, satisfaction, and the PCC. *Linguistic Inquiry* 55(1), 39-94. https://doi.org/10.1162/ling_a_00455.

Demuth, Katherine. 1990. Locatives, impersonals, and expletives in Sesotho. *The linguistic review* 7, 233-249. https://doi.org/10.1515/tlir.1990.7.3.233.

Demuth, Katherine & Mmusi, Sheila. 1997. Presentational focus and thematic structure in comparative Bantu. *Journal of African languages and linguistics* 18(1), 1-20. https://doi.org/10.1515/jall.1997.18.1.1.

Den Dikken, Marcel. 1995. Binding, expletives, and levels. *Linguistic Inquiry* 26(2), 347-354.

Diercks, Michael. 2012. Parameterizing Case: Evidence from Bantu. *Syntax* 15(3), 253-286. https://doi.org/10.1111/j.1467-9612.2011.00165.x.

Diercks, Michael. 2017. Locative inversion. In Everaert, Martin & van Riemsdijk, Henk (eds.) *The Wiley Blackwell companion to syntax.* http://doi.org/10.1002/9781118358733.wbsyncom082.

Doeleman, Toby. 1998. Quotative inversion in French. *Cornell working papers in linguistics* 16, 67-89.

Doggett, Teal Bissell. 2004. *All Things Being Unequal: Locality in Movement*. Doctoral dissertation. MIT.

Downing, Bruce. 1969. Vocatives and third-person imperatives in English. *Papers in linguistics* 1, 570-592.

Etxepare, Ricardo. 2008. On quotative constructions in Iberian Spanish. *Typological studies in language* 35-77. https://doi.org/10.1075/tsl.80.04etx.

Fernández-Serrano, Irene. 2022. *Agreement and optionality: Evidence from Spanish variation*. Doctoral dissertation. Universitat Autònoma de Barcelona.

Gärtner, Hans-Martin & Gyuris, Beáta. 2014. A note on quotative inversion in Hungarian. *Finno-Ugric languages and linguistics* 3(1-2), 2-30.

Georgi, Doreen. 2013. A relativized probing approach to person encoding in local scenarios. *Linguistic Variation* 12(2), 153-210. https://doi.org/10.1075/lv.12.2.02geo.

Gotah, Selikem. 2024. Deriving Ewe (Tongugbe) *nyá*-constructions. *Linguistic Variation*, 201-224. https://doi.org/10.1075/lv.23061.got.

Grohmann, Kleanthes. 2003. *Prolific domains: On the anti-locality of movement dependencies*. John Benjamins. https://doi.org/10.1075/la.66

Gruber, Jeffrey. 1997. Modularity in a Configurational Theta Theory. In DiSciullo, Anna-Maria (ed.) *Projections and interface conditions: Essays on modularity*, 155-200. https://doi.org/10.1093/oso/9780195104141.003.0007.

Gruber, Jeffrey & Collins, Chris. 1997. Argument projection, thematic configurationality, and Case theory. In DiSciullo, Anna-Maria (ed.) *Projections and interface conditions: Essays on modularity*, 130-154.

Haegeman, Liliane. 2013. The syntax of registers: Diary subject omission and the privilege of the root. *Lingua* 130, 88-110. https://doi.org/10.1016/j.lingua.2013.01.005.

Haegeman, Liliane & Ihsane, Tabea. 1999. Subject ellipsis in embedded clauses in English. *English language and linguistics* 3(1), 117-145. https://doi.org/10.1017/S1360674399000155.

Hale, Ken & Keyser, Samuel Jay. 2002. *Prolegomenon to a Theory of Argument Structure*. Cambridge, MA: MIT Press.

Halpert, Claire. 2012. Structural case and the nature of vP in Zulu. In *Proceedings of NELS 42*.

Halpert, Claire. 2015. *Argument licensing and agreement*. Oxford University Press. https://doi.org/10.1093/acprof:oso/9780190256470.001.0001.

Halpert, Claire. 2017. Prosody/syntax mismatches in the Zulu conjoint/disjoint alternation. In Hyman, Larry & van der Wal, Jenneke (eds.) *The Conjoint/Disjoint Alternation in Bantu*, 329-349. https://doi.org/10.1515/9783110490831-012.

Halpert, Claire. 2019. Raising, unphased. *Natural Language & Linguistic Theory* 37, 123–165. https://doi.org/10.1007/s11049-018-9407-2.

Halpert, Claire & Fong, Suzana. 2023. A-dependencies. Manuscript. Lingbuzz.

Halpert, Claire & Zeijlstra, Hedde. 2024. Off phases: It's all relative(ized). Manscript. University of Minnesota & Georg-August-Universität Göttingen. Lingbuzz.

Heim, Irene & Kratzer, Angelika. 1998. *Semantics in Generative Grammar*. Malden, MA: Blackwell Publishers.

Kasenov, Daniar. 2025. *External argument diagnostics expose variable argument linking in Russian unergatives and transitives*. Manuscript. New York University.

Ke, Hezao. 2019. *The Syntax, Semantics and Processing of Agreement and Binding Grammatical Illusions*. Doctoral dissertation. University of Michigan.

Keine, Stefan & Zeijlstra, Hedde. 2024. Clause-internal successive cyclicity: phasality or DP intervention? Manuscript. UCLA and Georg-August-Universität Göttingen. Lingbuzz.

Kilgarrif, Adam & Grefenstette, Gregory, 2003. Introduction to the special issue on the Web as corpus. *Computational Linguistics* 29(3), 333-347.

Koopman, Hilda & Szabolcsi, Anna. 2000. *Verbal Complexes*. MIT Press.

Koopman, Hilda. 1991. The Verb Particle Construction and the Syntax of PPs. Manuscript. UCLA.

Kratzer, Angelika. 1996. Severing the external argument from its verb. In Rooryck, Johan et al. (eds.) *Phrase structure and the lexicon*, 109–137. Springer.

Kusmer, Leland. 2019. Prosody and the conjoint/disjoint alternation in Tshivenḓa. In Clem, Emily & Jenks, Peter & Sande, Hannah (eds.) *Theory and description in African linguistics*, 213-229. https://doi.org/10.5281/zenodo.3367142.

Lee, Seunghun & Hlungwani, Crous. 2015. Aspectual auxiliary verbs in Xitsonga. *Stellenbosch papers in linguistics plus* 48, 113-135. http://doi.org/10.5842/48-0-678.

Lee-Goldman, Russell. 2011. *Context in Constructions*. Doctoral dissertation. UC Berkeley.

Levine, Robert. 1989. On focus inversion: Syntactic valence and the role of a SUBCAT list. *ZAS papers in linguistics* 43:97-122.

Mallén, E. 1991. A syntactic analysis of secondary predication in Spanish. *Journal of Linguistics, 27*(2), 375-403. https://doi.org/10.1017/S002222670001272X.

Marten, Lutz. 2006. Locative inversion in Otjiherero: More on morphosyntactic variation in Bantu. *ZAS papers in linguistics* 43, 97-122.

Marten, Lutz & van der Wal, Jenneke. 2015. A typology of Bantu subject inversion. *Linguistic Variation* 14(2), 318-336.

Meeusen, Achille. 1959. *Essai de grammaire rundi*. Tervuren.

Merchant, Jason. 2013. Voice and ellipsis. *Linguistic Inquiry* 44(1), 77–108. http://doi.org/10.1162/LING_a_00120.

Milway, Daniel. 2023. A formalization of Agree as a derivational operation. *Biolinguistics* 17, e9877. https://doi.org/10.5964/bioling.9877.

Murphy, Andrew. 2022. Parasitic gaps diagnose A-movement in quotative and locative inversion. *Snippets* 43, 4-6. http://dx.doi.org/10.7358/snip-2022-043-murp.

Myler, Neil. 2013. On coming the pub in the North West of England: Accusative unaccusatives, dependent case, and preposition incorporation. Journal of Comparative Germanic Linguistics 16(2–3), 189–207. https://doi.org/10.1007/s10828-013-9055-1.

Ndayiragije, Juvénal. 1999. Checking economy. *Linguistic Inquiry* 30, 399–444. https://doi.org/10.1162/002438999554129.

Neumann, Polina. 2025. *Diagnosing small nominals: Theoretical implications from Moksha and Hill Mari*. Doctoral dissertation. University of Maryland.

Nevins, Andrew. 2007. The Representation of Third Person and Its Consequences for Person-Case Effects. *Natural Language & Linguistic Theory* 25, 273-313. http://doi.org/10.1007/s11049-006-9017-2.

Nevins, Andrew. 2011. Multiple agree with clitics: person complementarity vs. omnivorous number. *Natural Language and Linguistic Theory* 29, 939–971. https://doi.org/10.1007/s11049-011-9150-4.

Nicolle, Steve. 2013. *A grammar of Digo*. Dallas, TX: SIL International.

Oxford, Will. 2023. A tale of two inverses. *Syntax* 26, 311-354. http://doi.org/10.1111/synt.12251

Pereltsvaig, Asya. 2006. Small nominals. *Natural language & linguistic theory* 24, 433-500.

Pesetsky, David. 1989. Language-particular processes and the Earliness Principle. Manuscript. MIT.

Pietraszko, Asia. 2020. The coming apart of case and focus in Bantu. *Natural Language & Linguistic Theory* 39(2), 579-599. https://doi.org/10.1007/s11049-020-09481-z.

Pietraszko, Asia. 2023. Timing-driven derivation of a NOM/ACC agreement pattern. *Glossa* 8(1). https://doi.org/10.16995/glossa.5849.

Pietraszko, Asia. 2024. *In-situ subjects in Ndebele reveal a clause-internal phase*. Presented May 2 2024 at the 55th Annual Conference on African Linguistics, Montreal, QC.

Pietraszko, Asia. 2025. In Defense of the Clause-Internal Phase: Evidence from Operational Opacity in Ndebele. Manuscript. Lingbuzz.

Preminger, Omer. 2014. *Agreement and its failures*. Cambridge, MA: MIT Press.

Preminger, Omer. 2019. What the PCC tells us about "abstract" agreement, head movement, and locality. *Glossa* 4(1), 13.

Pylkkänen, Liina. 2008. *Introducing Arguments*. Cambridge, MA: MIT Press.

Rezac, Milan. 2004. *Elements of cyclic syntax: Agree and Merge*. Doctoral dissertation. University of Toronto.

Richards, Norvin. 1999. Featural cyclicity and the ordering of multiple specifiers. In *Working minimalism*, 127-156. Cambridge, MA: MIT Press.

Rizzi, Luigi. 1990. *Relativized Minimality*. Cambridge, MA: MIT Press.

Rizzi, Luigi & Shlonsky, Ur. 2006. Satisfying the Subject Criterion by a Non Subject: English Locative Inversion and Heavy NP Shift. In Frascarelli, Maria (ed.) *Phases of Interpretation*, 341–361. Berlin: De Gruyter.

Roberts, Ian. 2010. *Agreement and Head Movement: Clitics, Incorporation, and Defective Goals*. Cambridge, MA: MIT Press.

Schütze, Carson. 2011. Linguistic evidence and grammatical theory. *WIREs Cognitive Science* 2, 206-221.

Shlonsky, Ur. 2024. From Bantu subject-object reversal to inverted copular sentences. *Rich Descriptions and Simple Explanations in Morphosyntax and Language Acquisition.* 438. Oxford University Press. https://doi.org/10.1093/oso/9780198889472.003.0017.

Staps, Camil & Rooryck, Johan. 2023. The interpretation of [+distal] in demonstratives and complementizers. *Linguistics* 61(5), 1195-1231. https://doi.org/10.1515/ling-2022-0178.

Stegovec, Adrian. 2024. The third way: Object reordering as ambiguous labeling resolution. *Linguistic review* 41(1), 187-223. https://doi.org/10.1515/tlr-2024-2007.

Storment, John David. 2025. *Projecting (your) voice: A theory of inversion and defective circumvention*. Doctoral dissertation. Stony Brook University.

Suñer, Margarita. 2000. The syntax of direct quotes with special reference to Spanish and English. *Natural Language & Linguistic Theory* 18, 525-578.

Titov, Elena. 2012. *Information Structure of argument order alternations*. Doctoral dissertation. University College London.

Titov, Elena. 2013. Scrambling and interfaces. In Karvovskaya, Lena et al. (eds.) *Information Structure: Empirical perspectives on theory*, 33-54. Potsdam: Universitatsverlag Potsdam.

Ura, Hiroyuki. 2000. On the definition of closeness in the mechanism of Agree. In *The proceedings of the 2nd international Asian GLOW conference*. 425-442.

Van der Spuy, Andrew. 1993. Dislocated noun phrases in Nguni. *Lingua* 90, 335-355.

Van der Wal, Jenneke. 2017. What is the conjoint/disjoint alternation? Parameters of crosslinguistic variation. In Hyman, Larry & van der Wal, Jenneke (eds.) *The Conjoint/Disjoint Alternation in Bantu*, 14-60. https://doi.org/10.1515/9783110490831-002.

Van der Wal, Jenneke & Hyman, Larry. 2017. *The Conjoint/Disjoint Alternation in Bantu.* De Gruyter Mouton. https://doi.org/10.1515/9783110490831.

Van Riemsdijk, Henk. 1978. *A case study in Syntactic Markedness: The binding nature of prepositional phrases*. Lisse: The Peder de Riddler Press.

Van Urk, Coppe. 2015. *A uniform syntax for phrasal movement: A case study of Dinka Bor*. Doctoral dissertation. MIT.

Weir, Andrew. 2008. *Subject pronoun drop in informal English.* MA thesis. University of Edinburgh.

Wikström, Peter. 2017. *I tweet like I talk: Aspects of speech and writing on Twitter*. Doctoral dissertation. Karlstad University.

Williams, Edwin. 1980. Predication. *Linguistic Inquiry* 11, 203-238.

Zeller, Jochen. 2013. Locative inversion in Bantu and predication. *Linguistics* 51(6) 1107-1146. https://doi.org/10.1515/ling-2013-0046.

Zeller, Jochen & Ngoboka, Jean Paul. 2018. Agreement with locatives in Kinyarwanda: A comparative analysis. *Journal of African languages and linguistics* 31(1), 65-106.

Zerbian, Sabine. 2006. Inversion structurers in Northern Sotho. *Southern African linguistics and applied language studies* 24(3), 361-376. https://doi.org/10.2989/16073610609486425.